

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

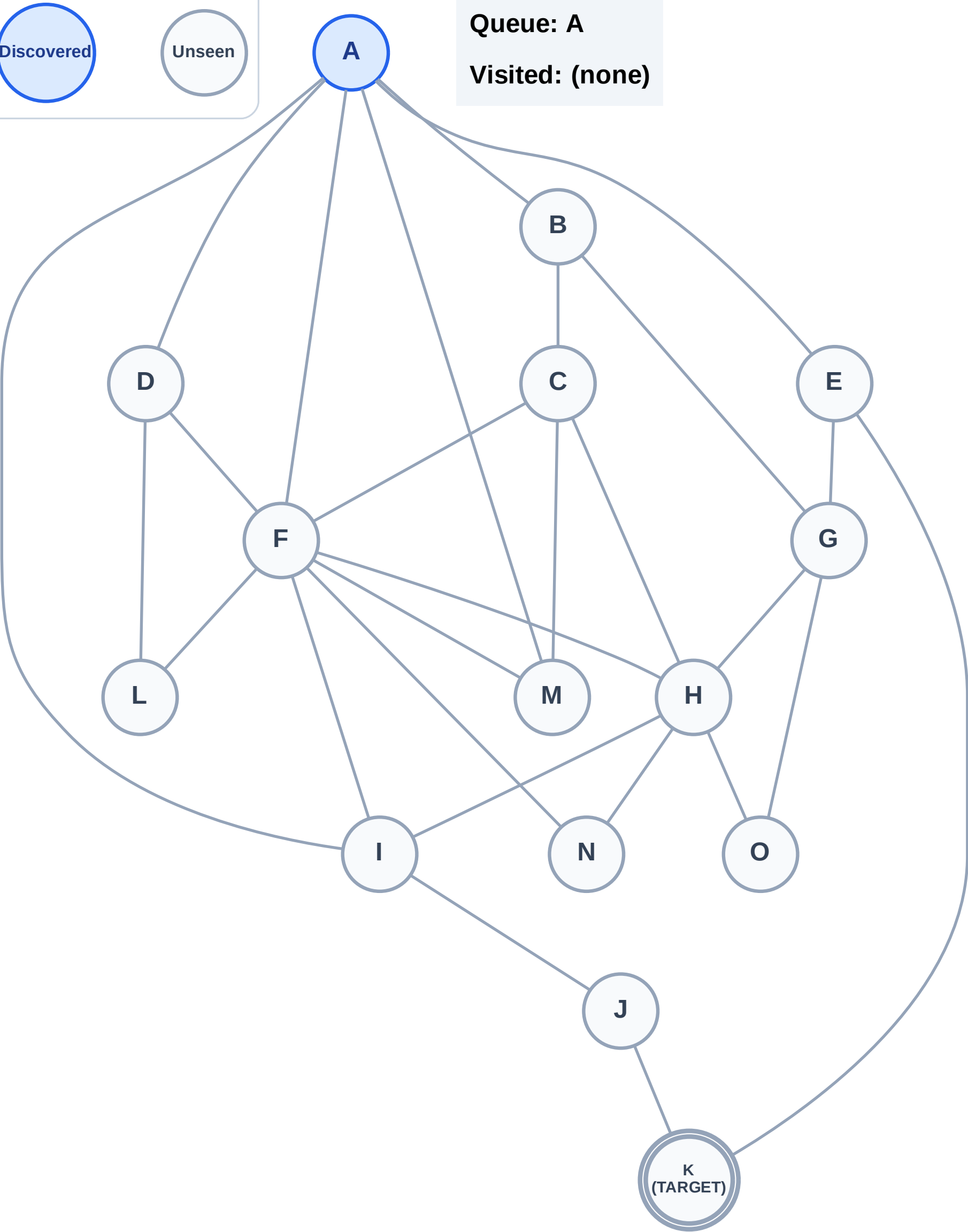
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

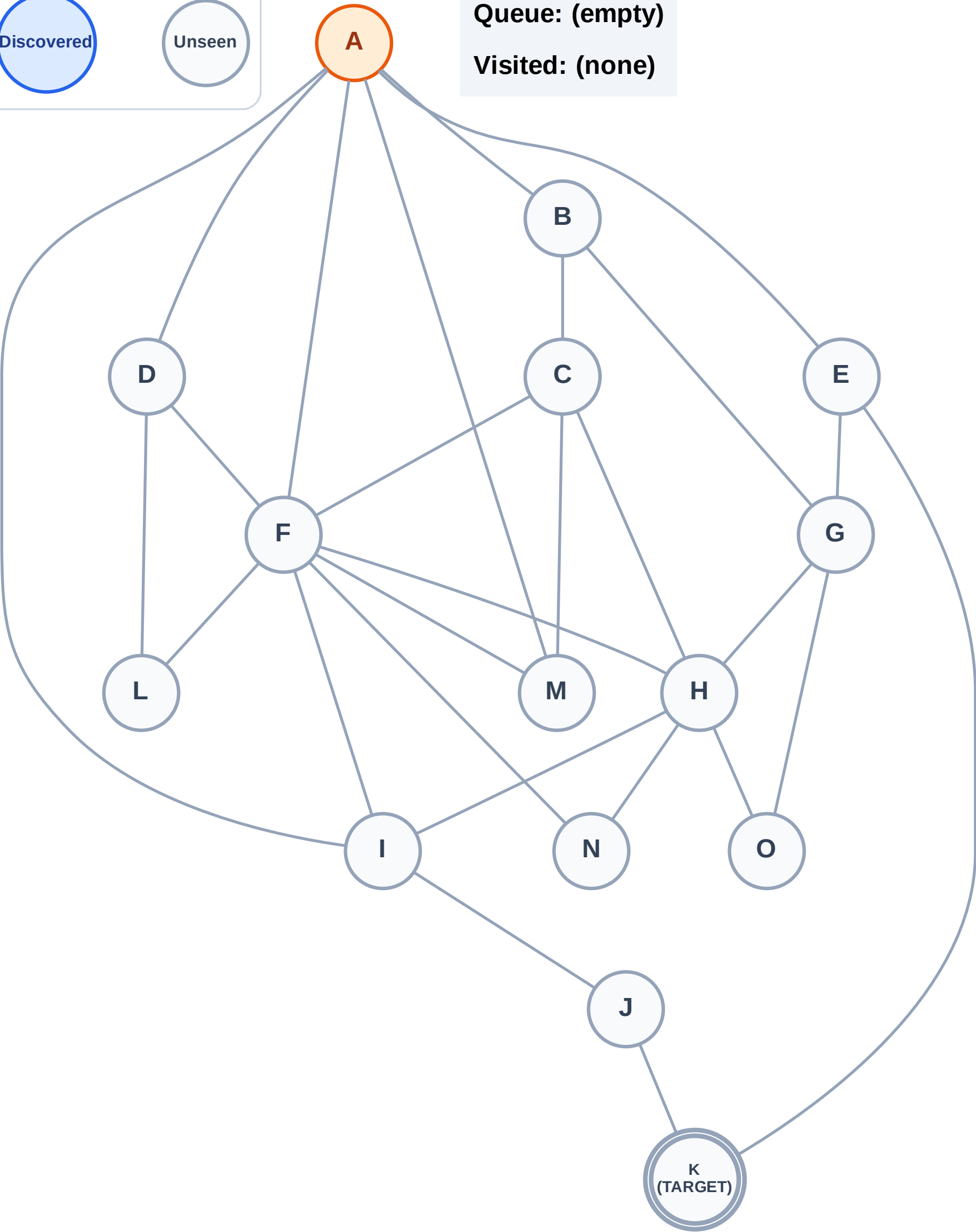
Complete

Processing

Discovered

Unseen

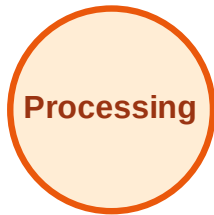
Queue: (empty)
Visited: (none)



BFS Traversal

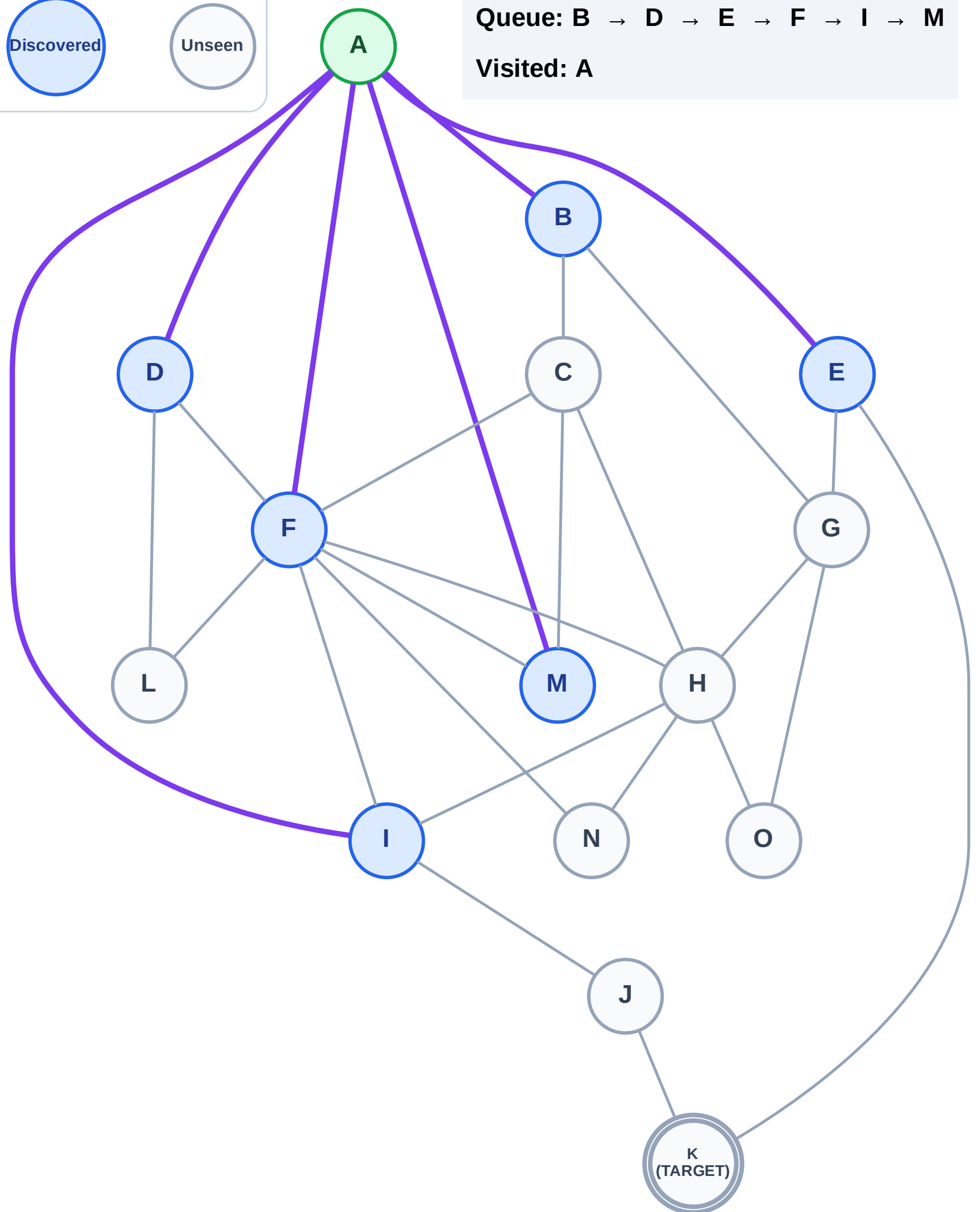
Step 3: Discover B, D, E, F, I, M from A

Legend



Queue: B → D → E → F → I → M

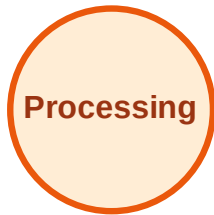
Visited: A



BFS Traversal

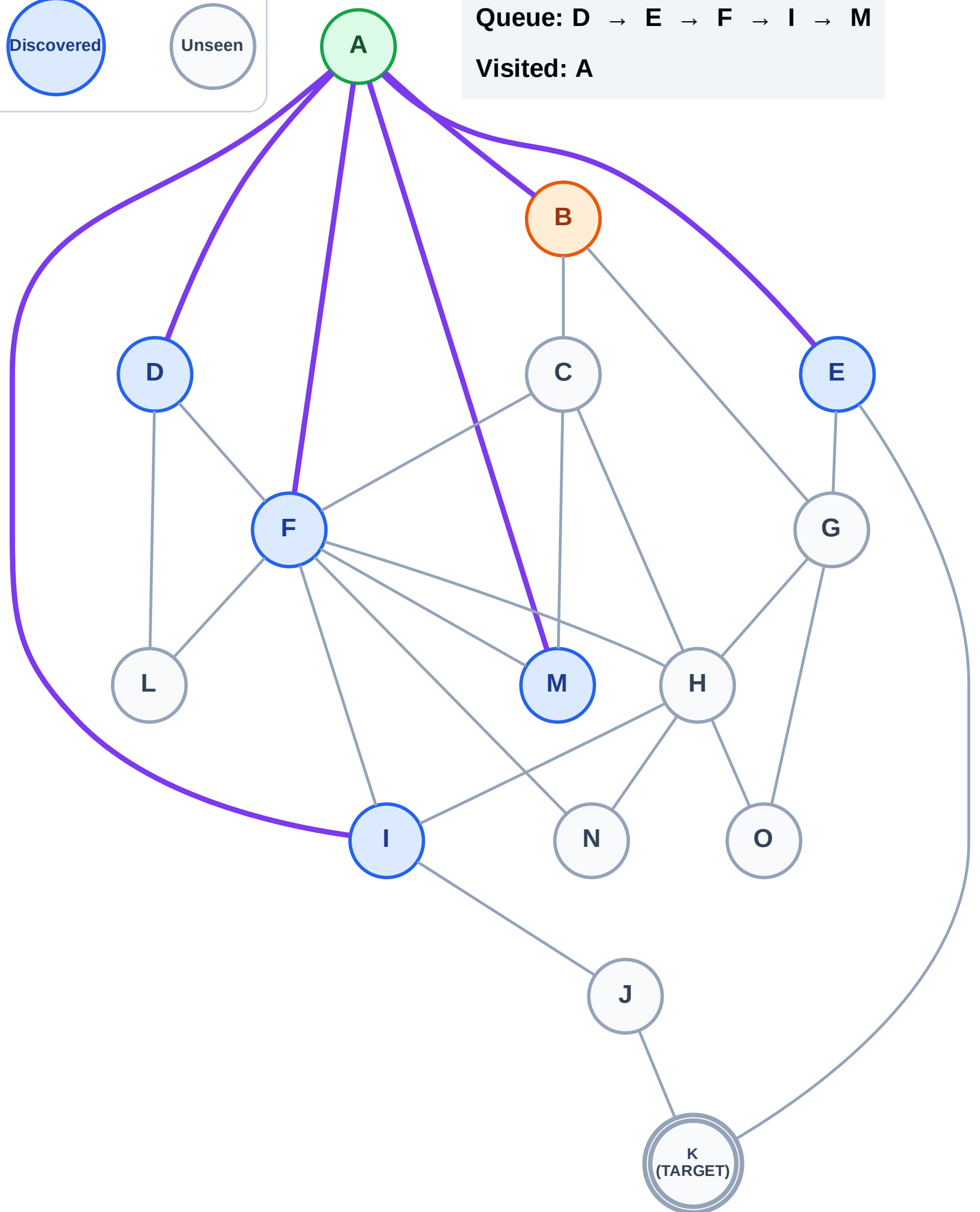
Step 4: Process node B

Legend



Queue: D → E → F → I → M

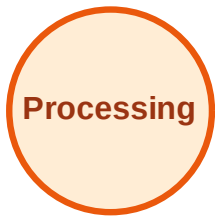
Visited: A



BFS Traversal

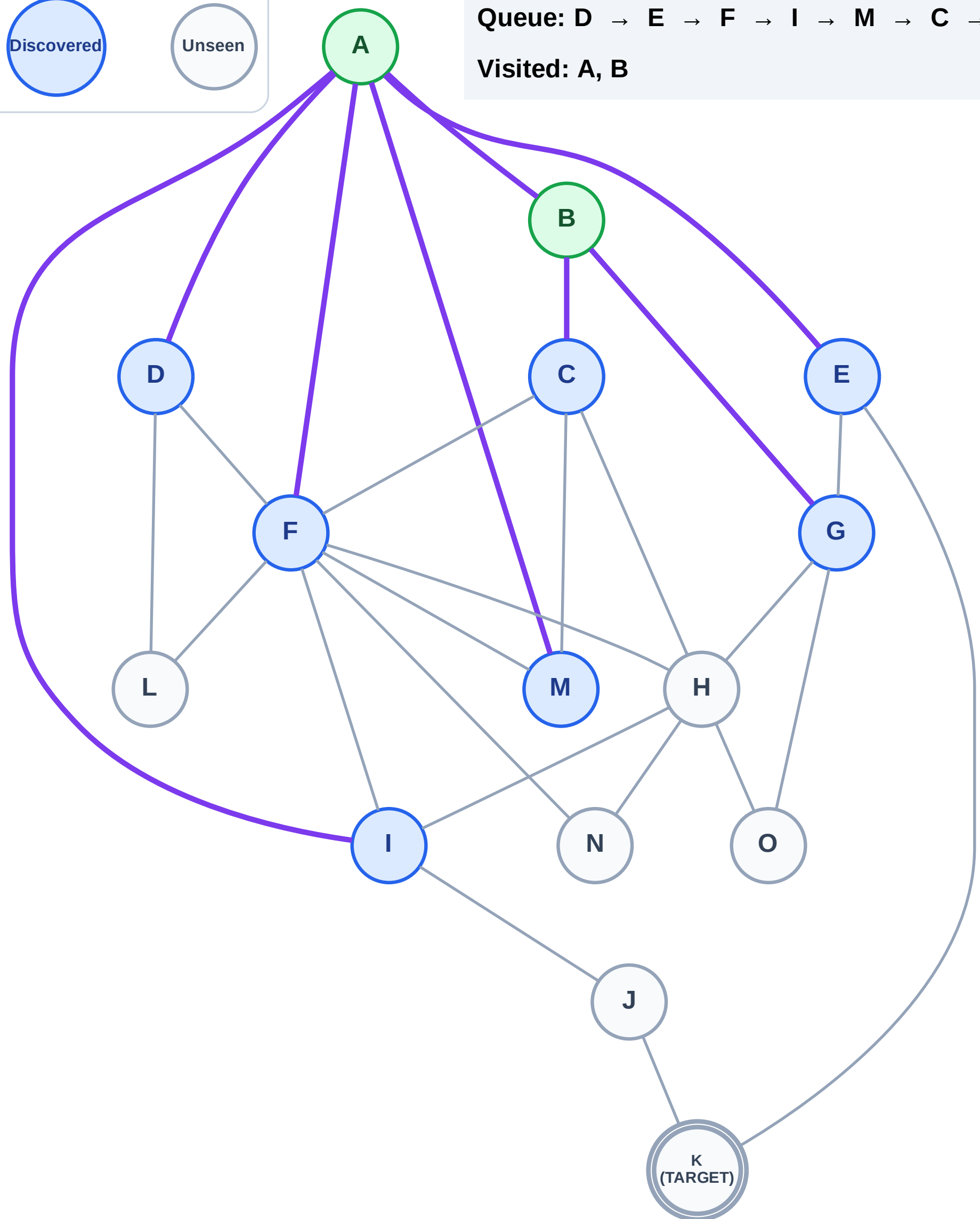
Step 5: Discover C, G from B

Legend



Queue: D → E → F → I → M → C → G

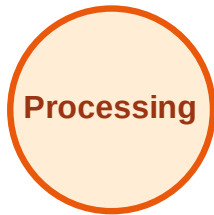
Visited: A, B



BFS Traversal

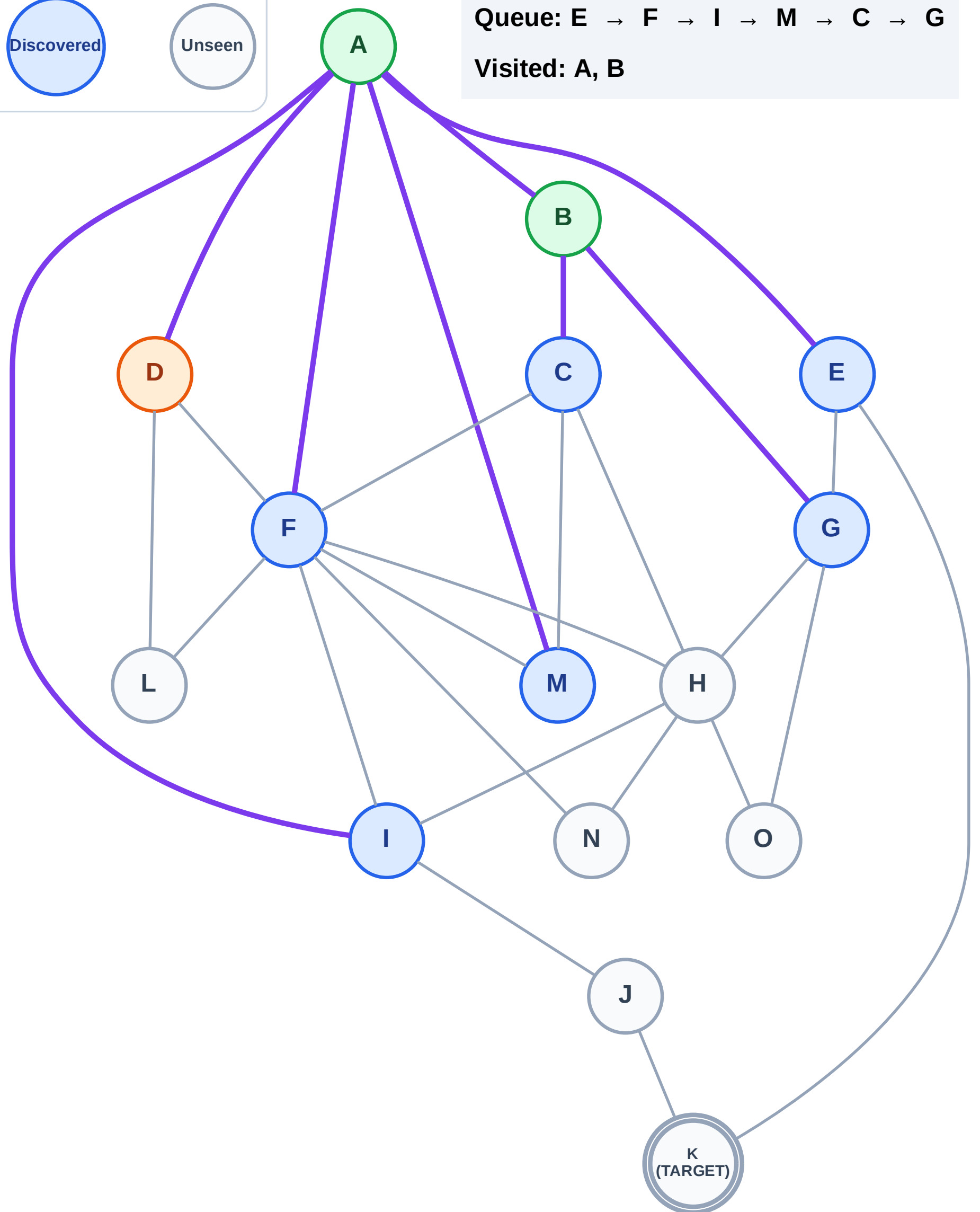
Step 6: Process node D

Legend

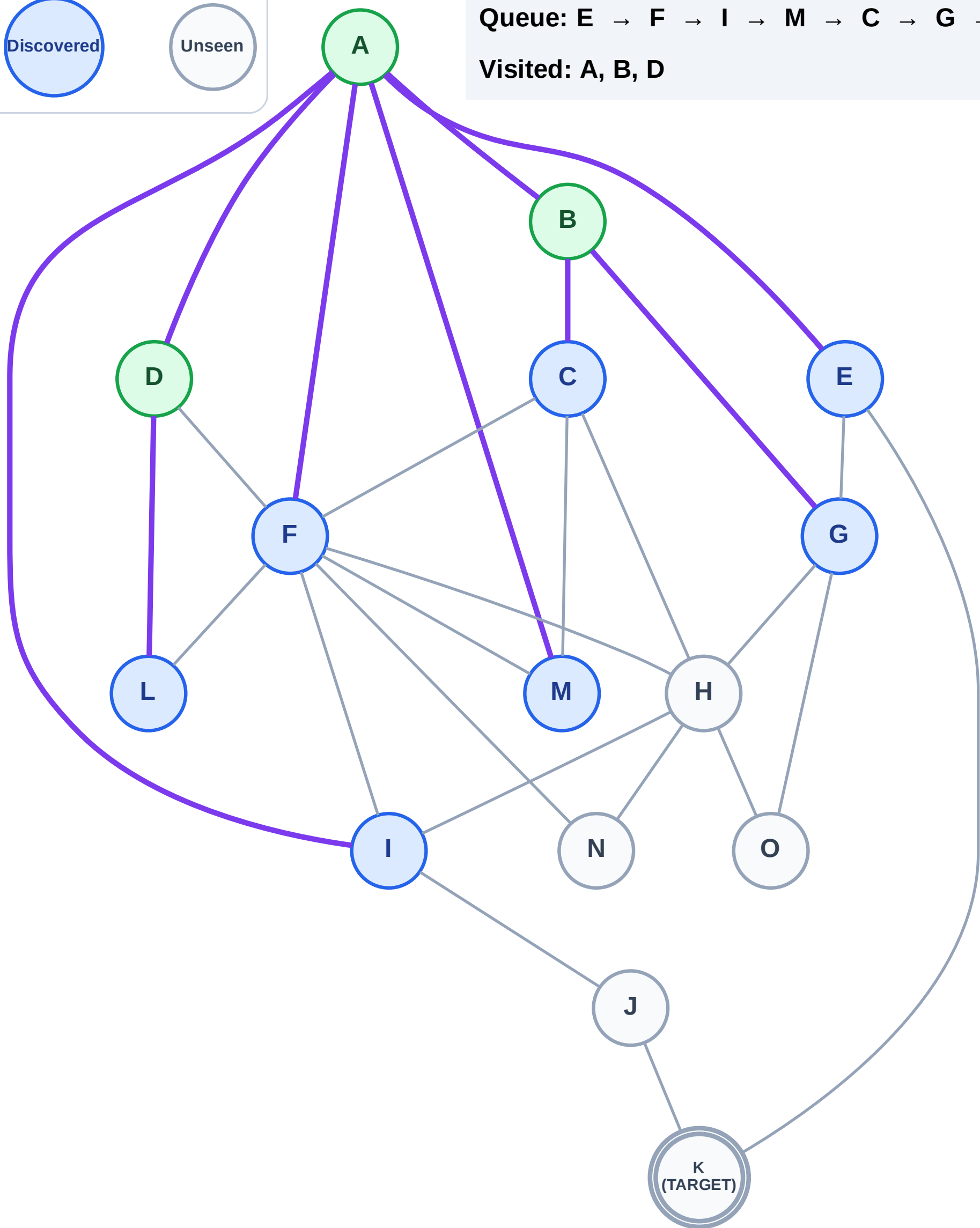


Queue: E → F → I → M → C → G

Visited: A, B



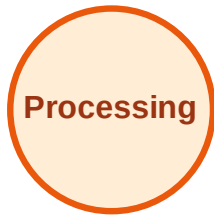
Step 7: Discover L from D



BFS Traversal

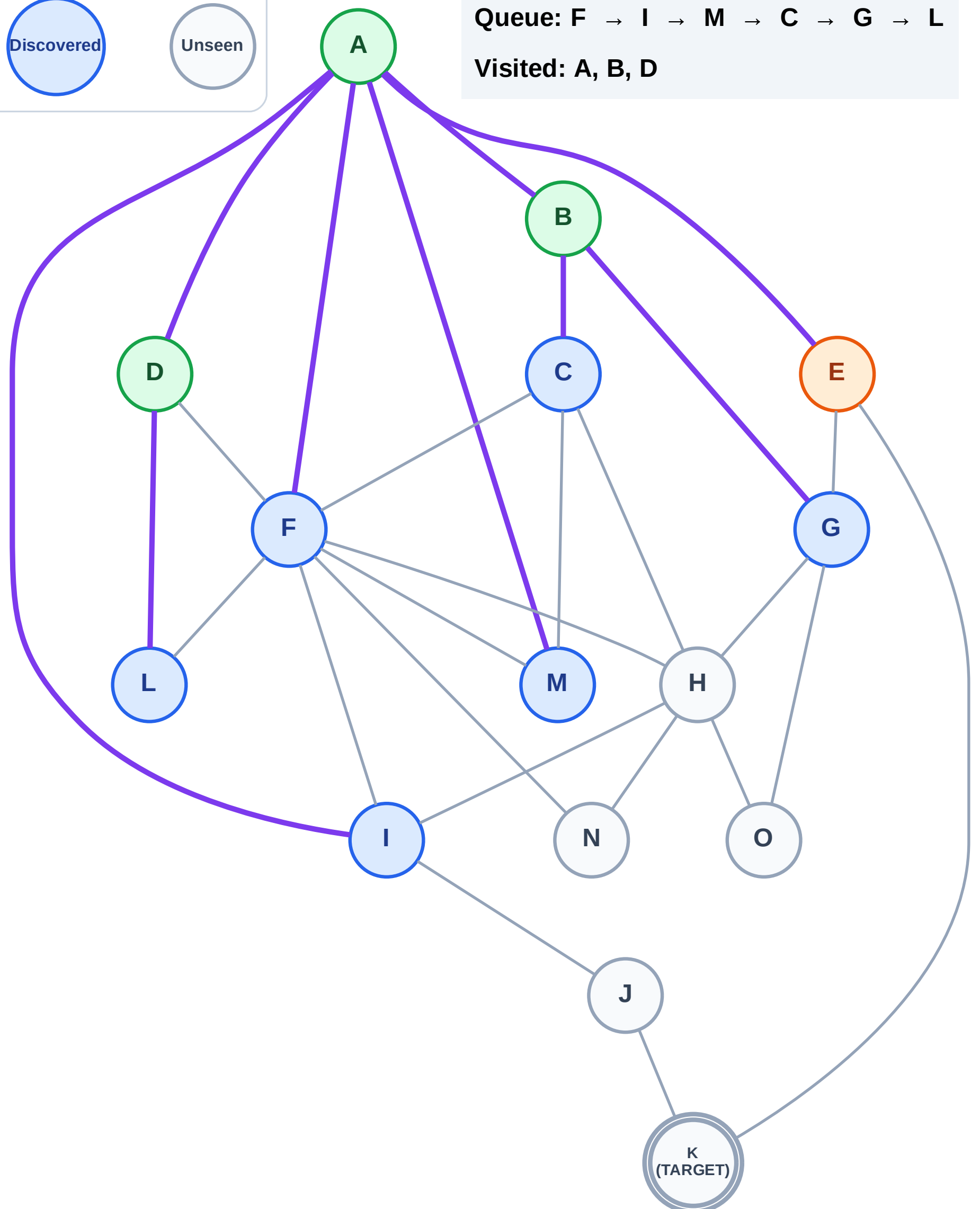
Step 8: Process node E

Legend



Queue: F → I → M → C → G → L

Visited: A, B, D



BFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → E → K

Legend

Complete

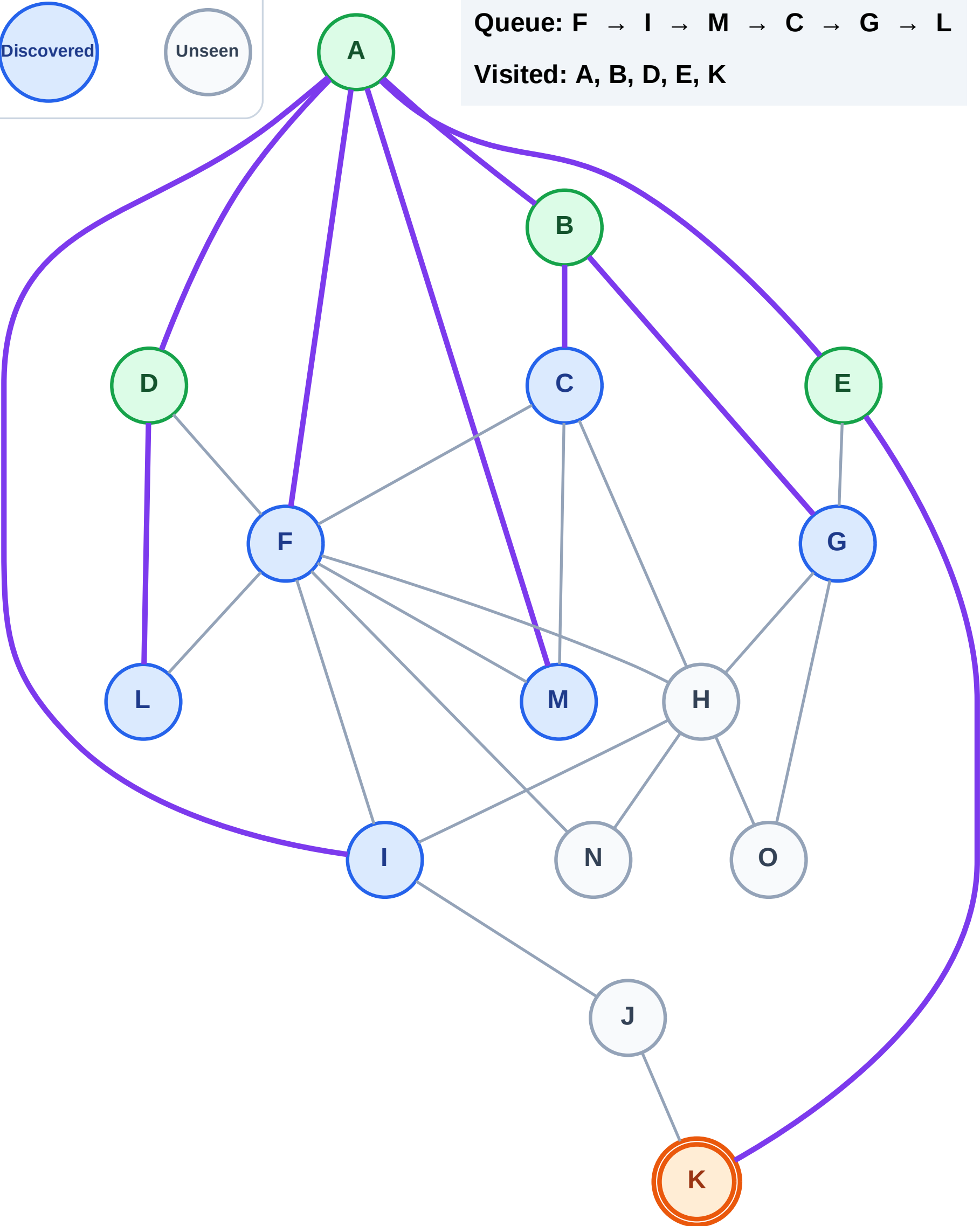
Processing

Discovered

Unseen

Queue: F → I → M → C → G → L

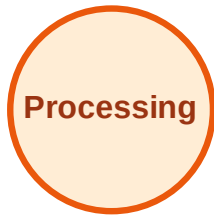
Visited: A, B, D, E, K



DFS Traversal

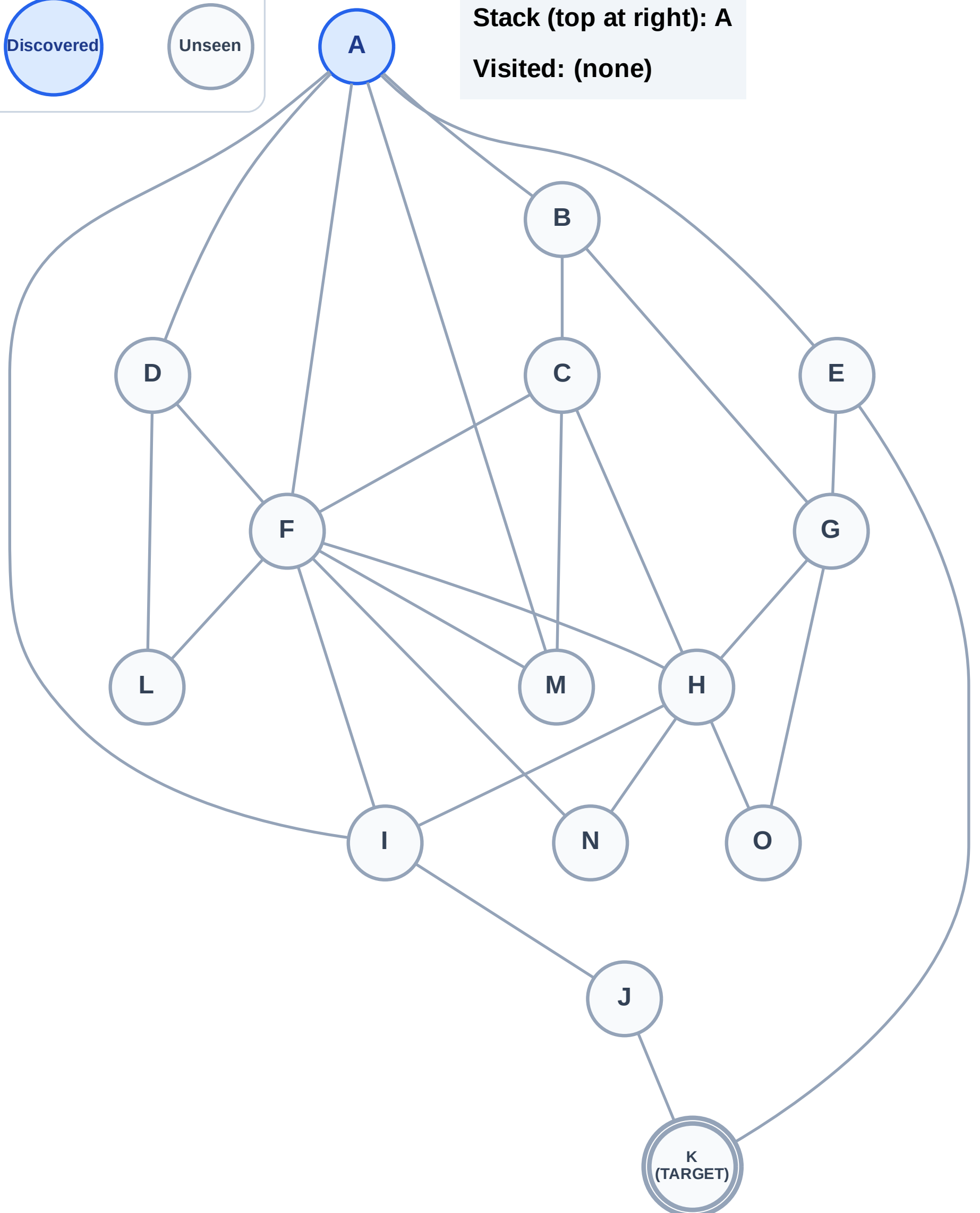
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

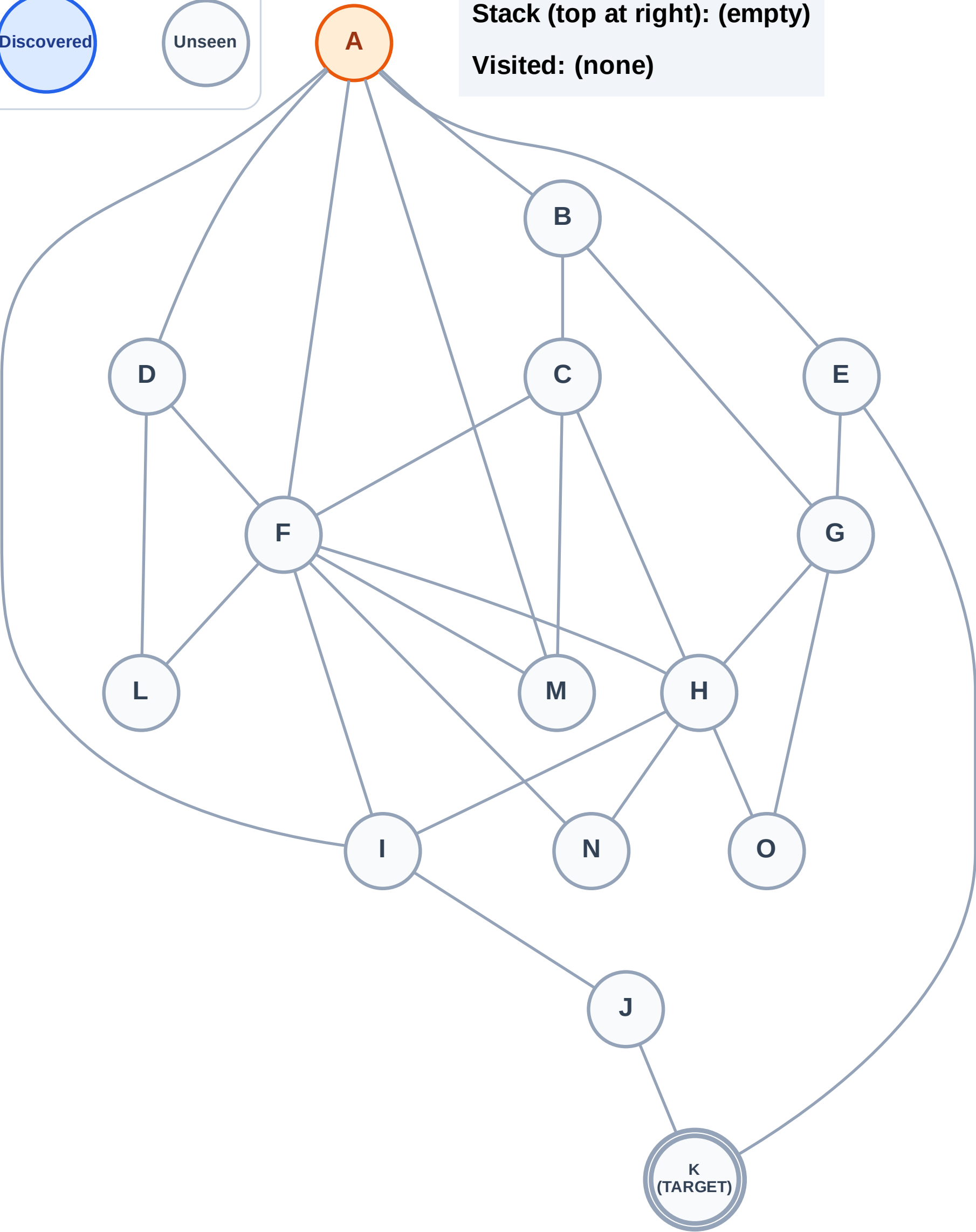
Complete

Processing

Discovered

Unseen

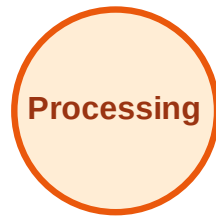
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

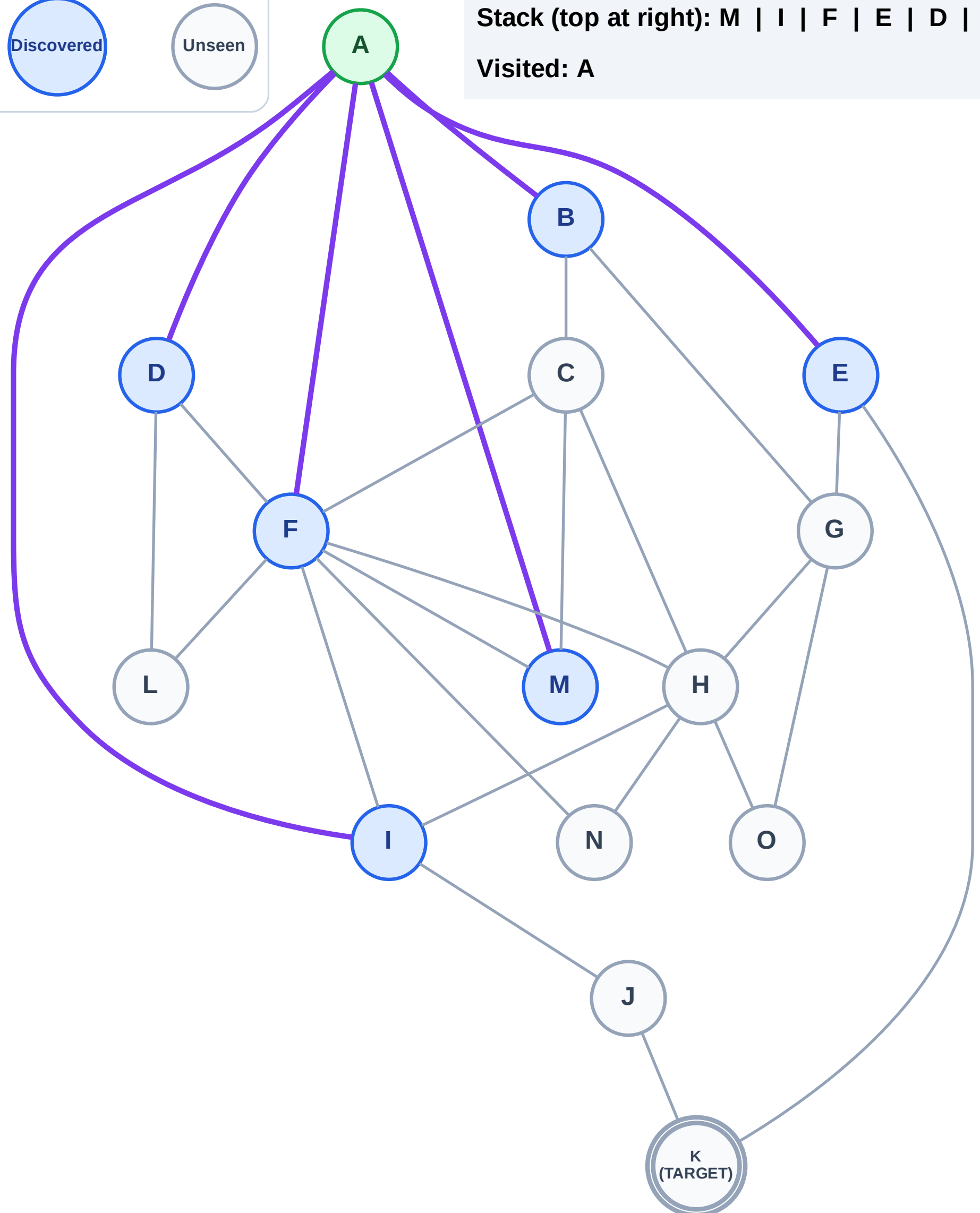
Step 3: Discover M, I, F, E, D, B from A

Legend



Stack (top at right): M | I | F | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

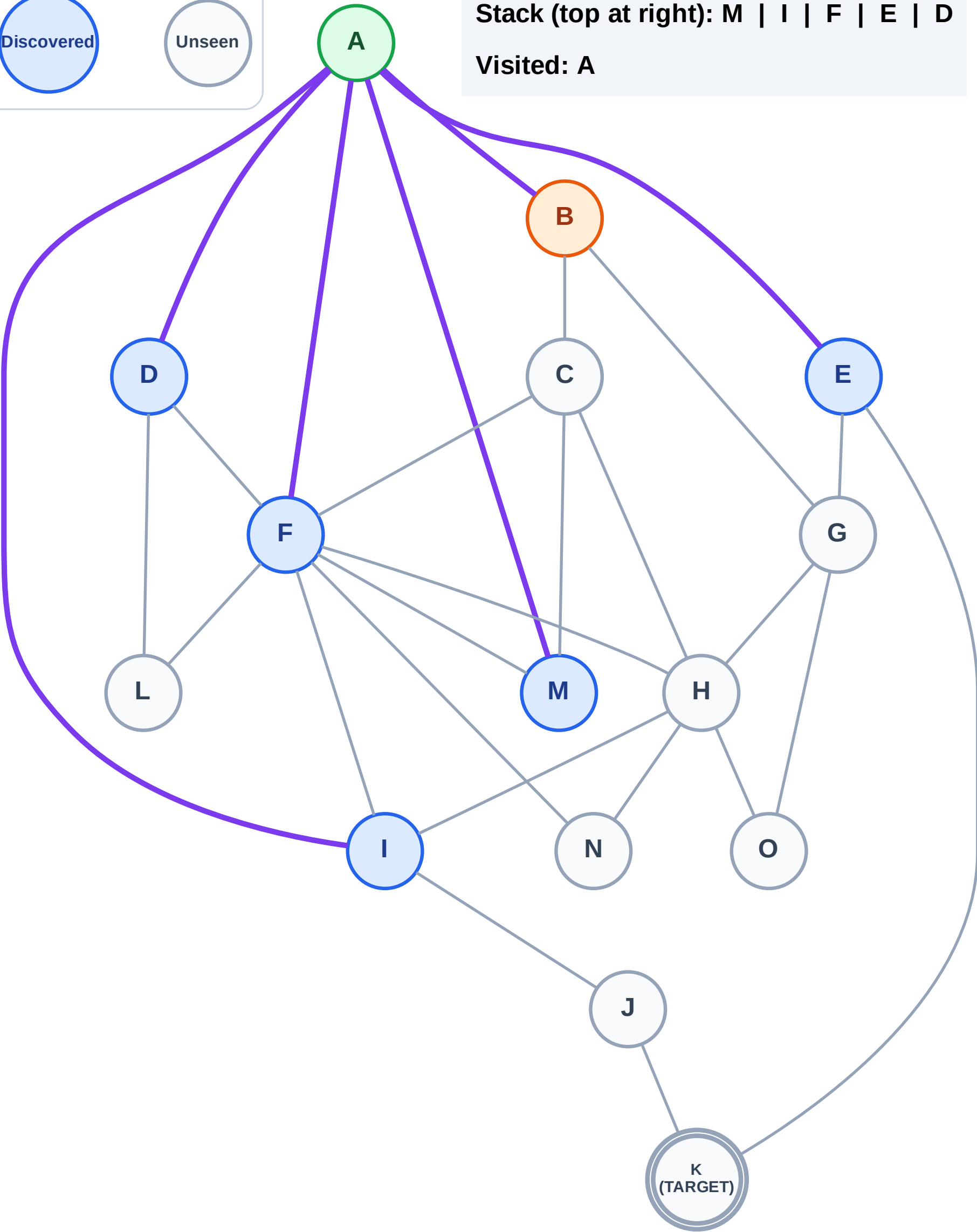
Processing

Discovered

Unseen

Stack (top at right): M | I | F | E | D

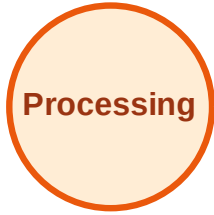
Visited: A



DFS Traversal

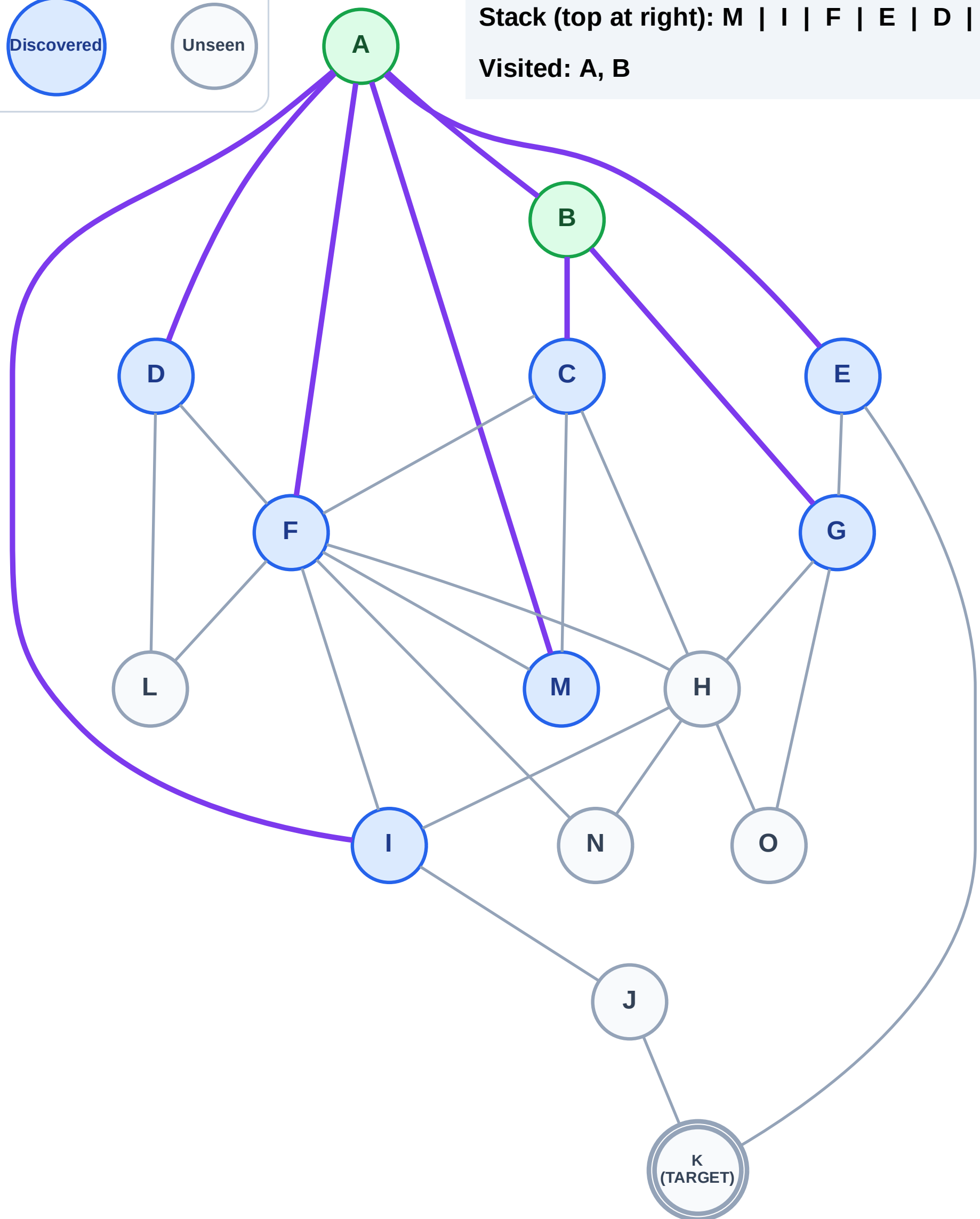
Step 5: Discover G, C from B

Legend



Stack (top at right): M | I | F | E | D | G | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

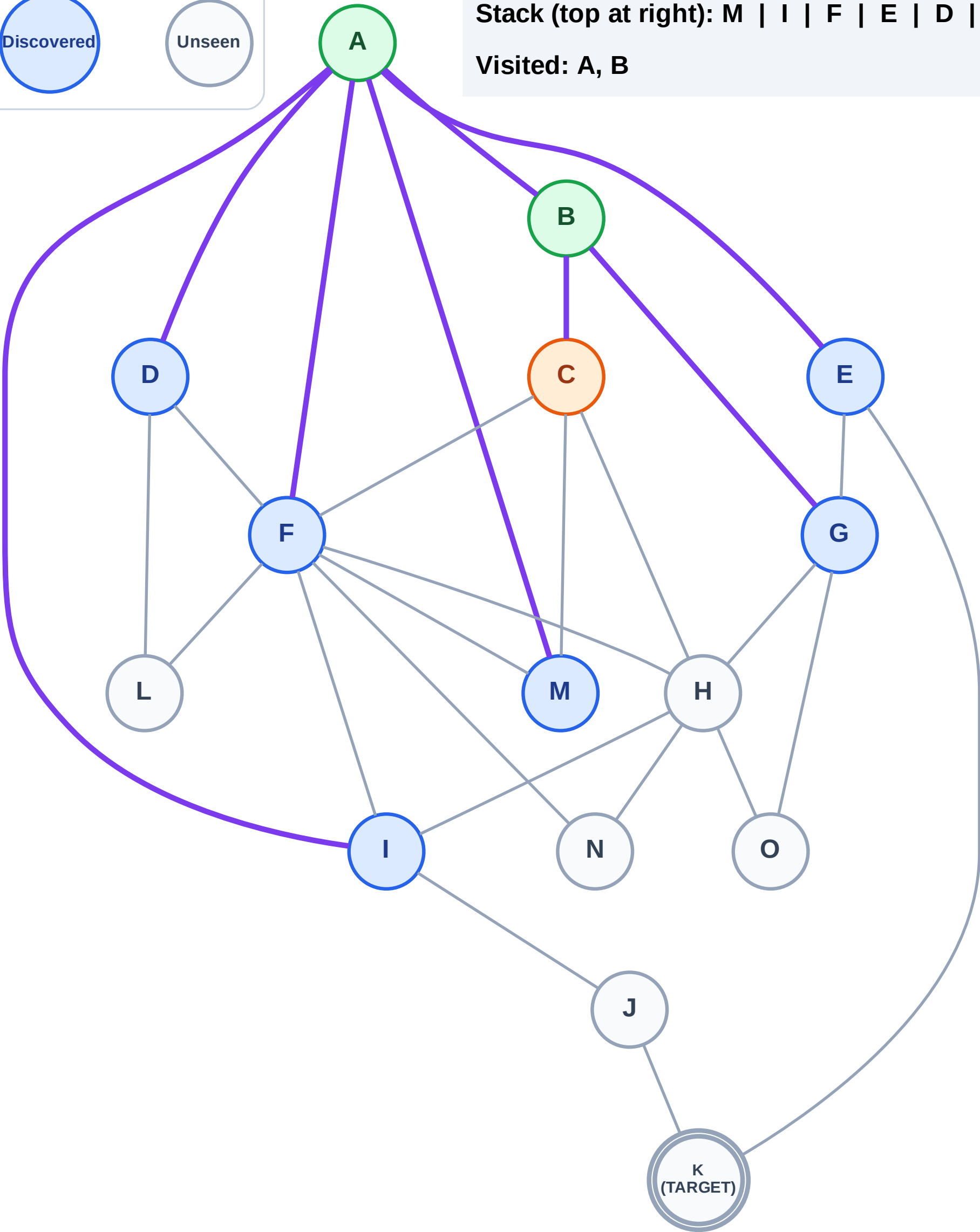
Processing

Discovered

Unseen

Stack (top at right): M | I | F | E | D | G

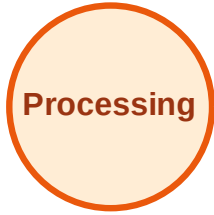
Visited: A, B



DFS Traversal

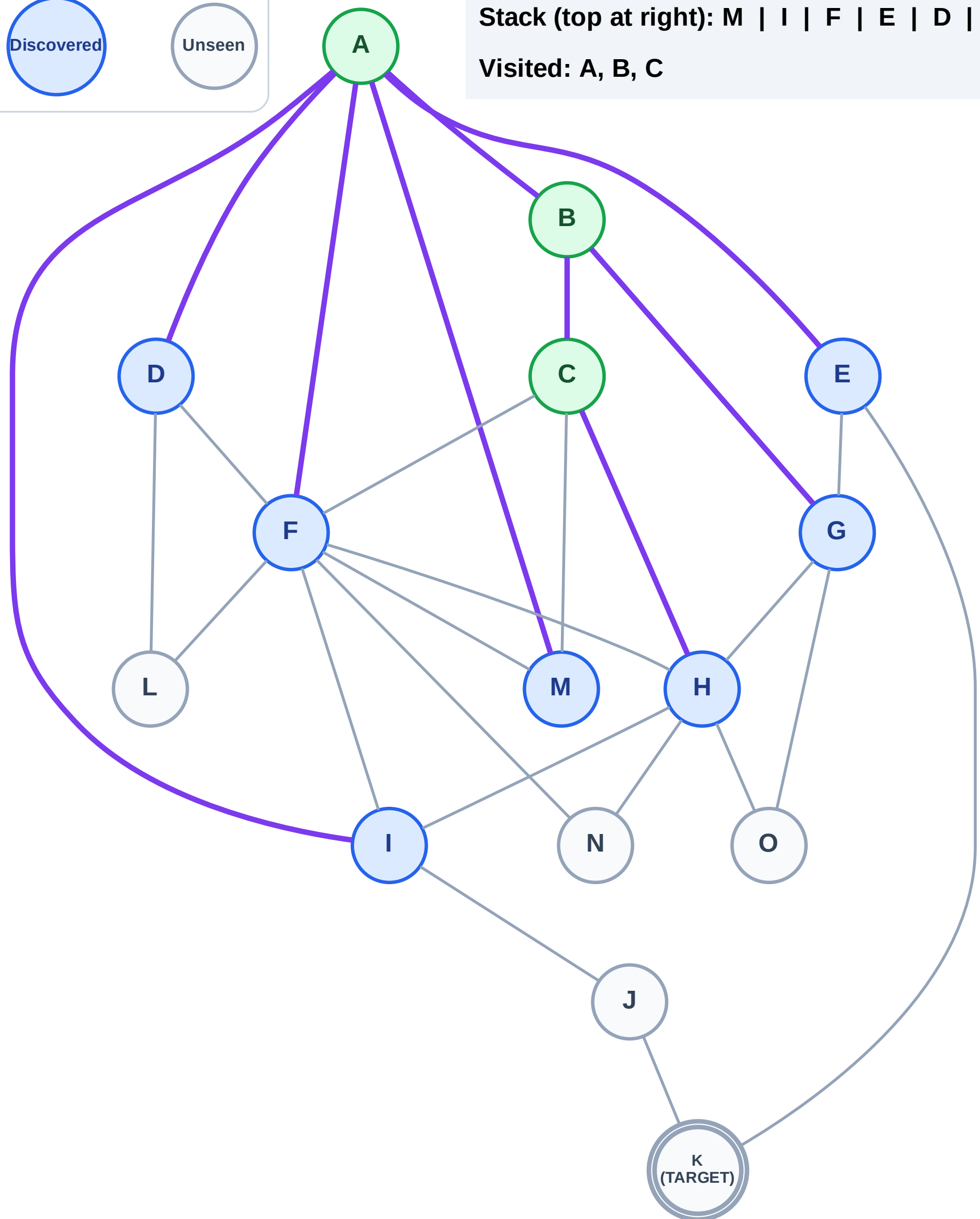
Step 7: Discover H from C

Legend



Stack (top at right): M | I | F | E | D | G | H

Visited: A, B, C



DFS Traversal

Step 8: Process node H

Legend

Complete

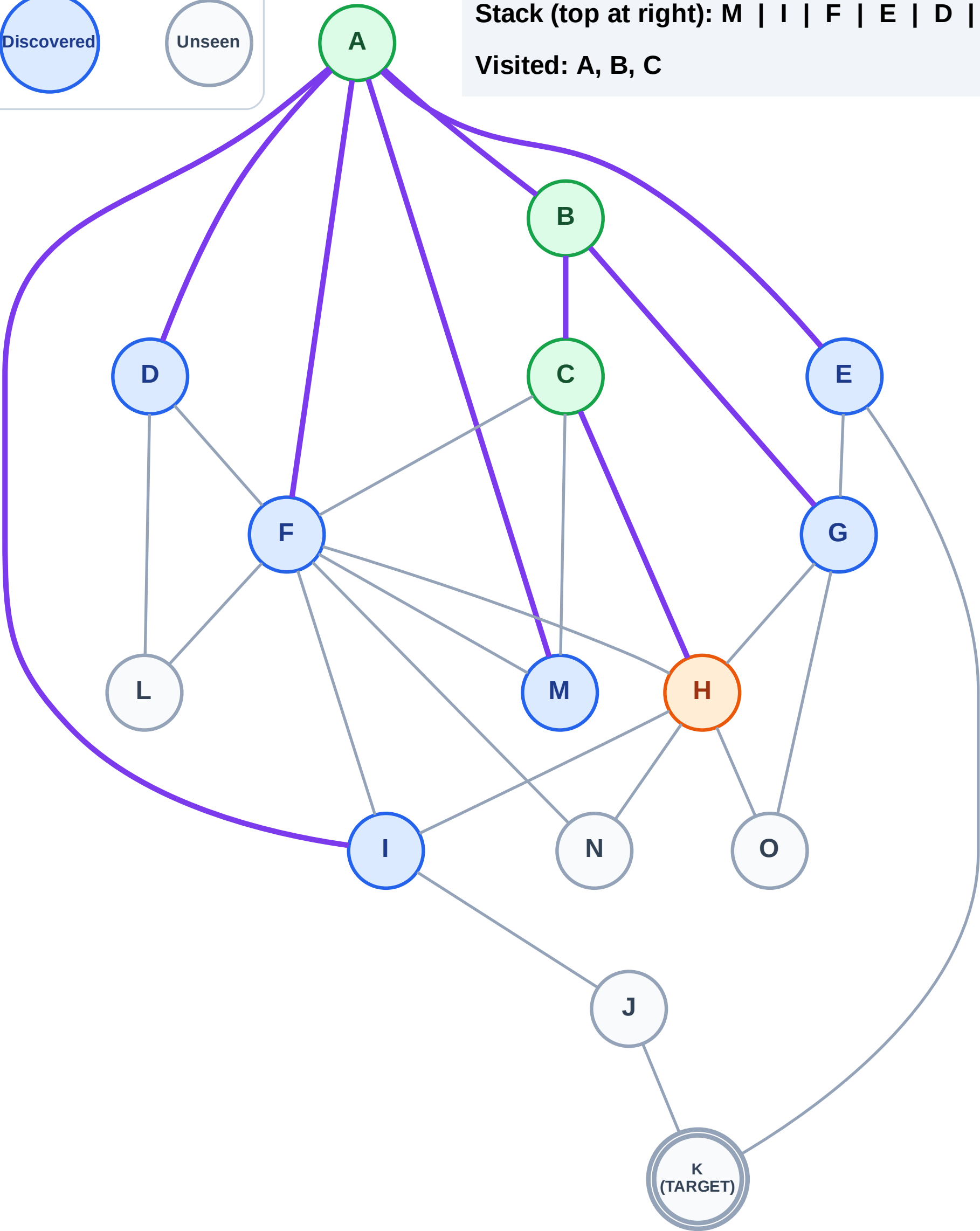
Processing

Discovered

Unseen

Stack (top at right): M | I | F | E | D | G

Visited: A, B, C



DFS Traversal

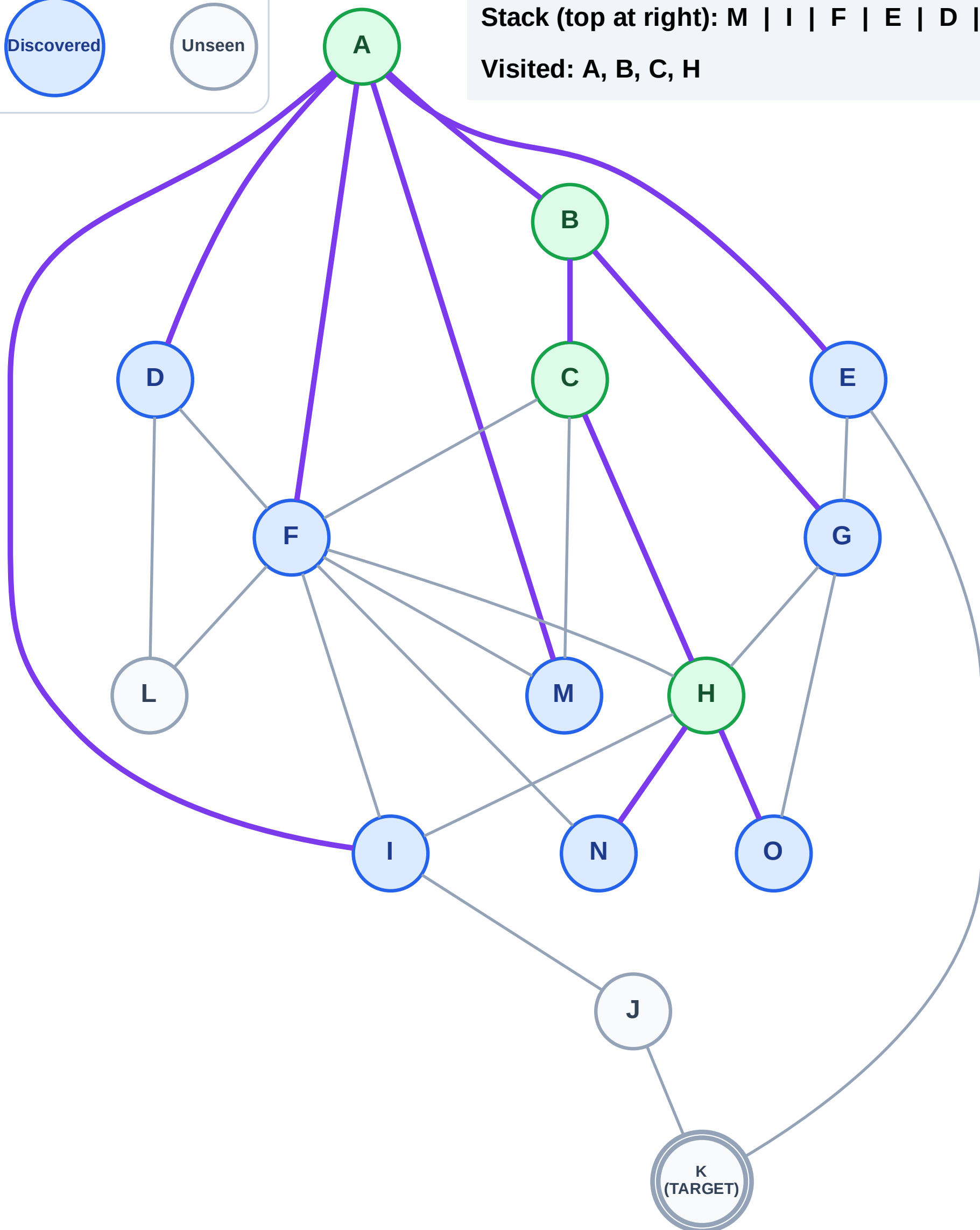
Step 9: Discover O, N from H

Legend



Stack (top at right): M | I | F | E | D | G | O | N

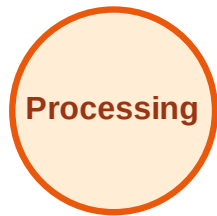
Visited: A, B, C, H



DFS Traversal

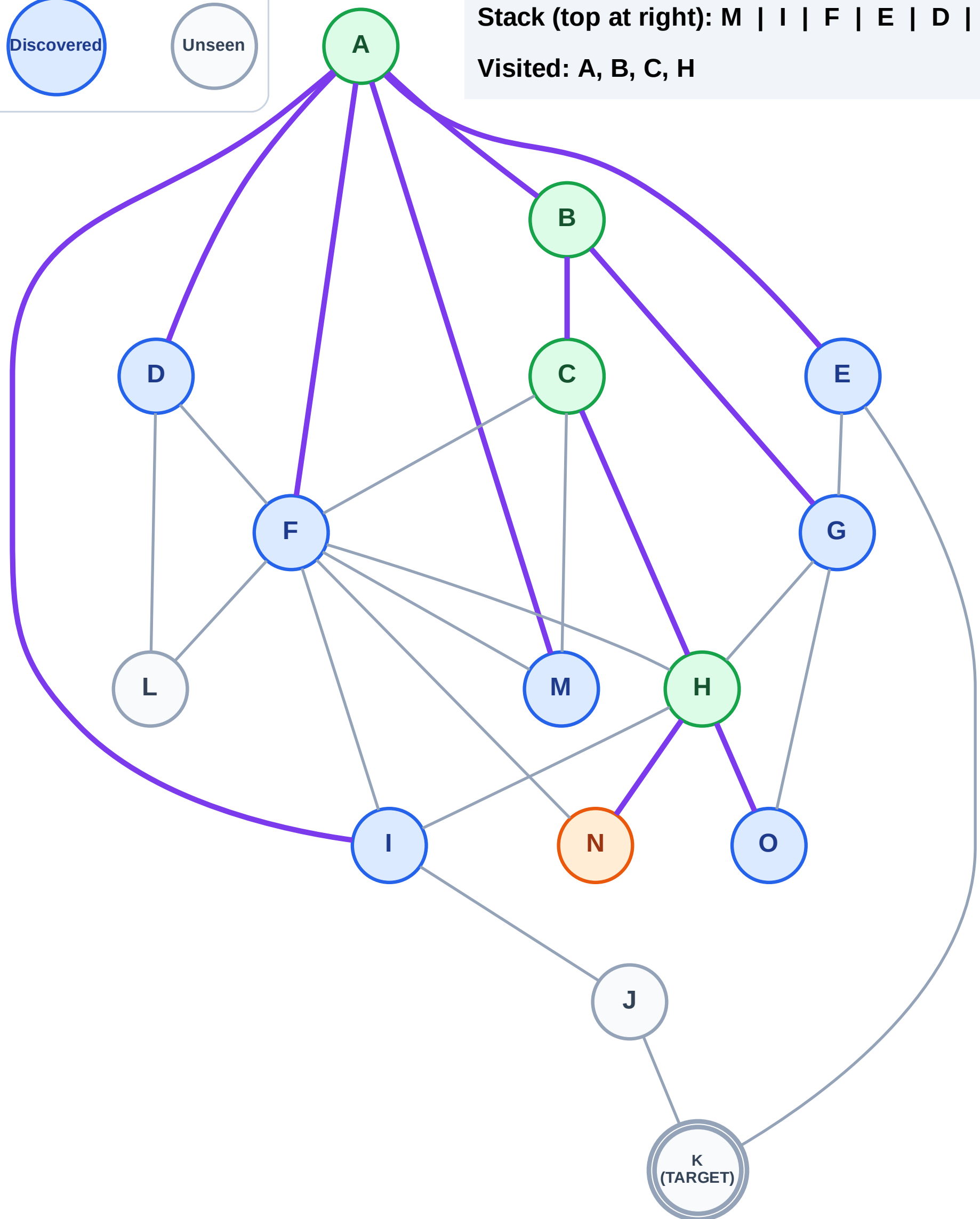
Step 10: Process node N

Legend



Stack (top at right): M | I | F | E | D | G | O

Visited: A, B, C, H



DFS Traversal

Step 11: No new neighbors from N

Legend

Complete

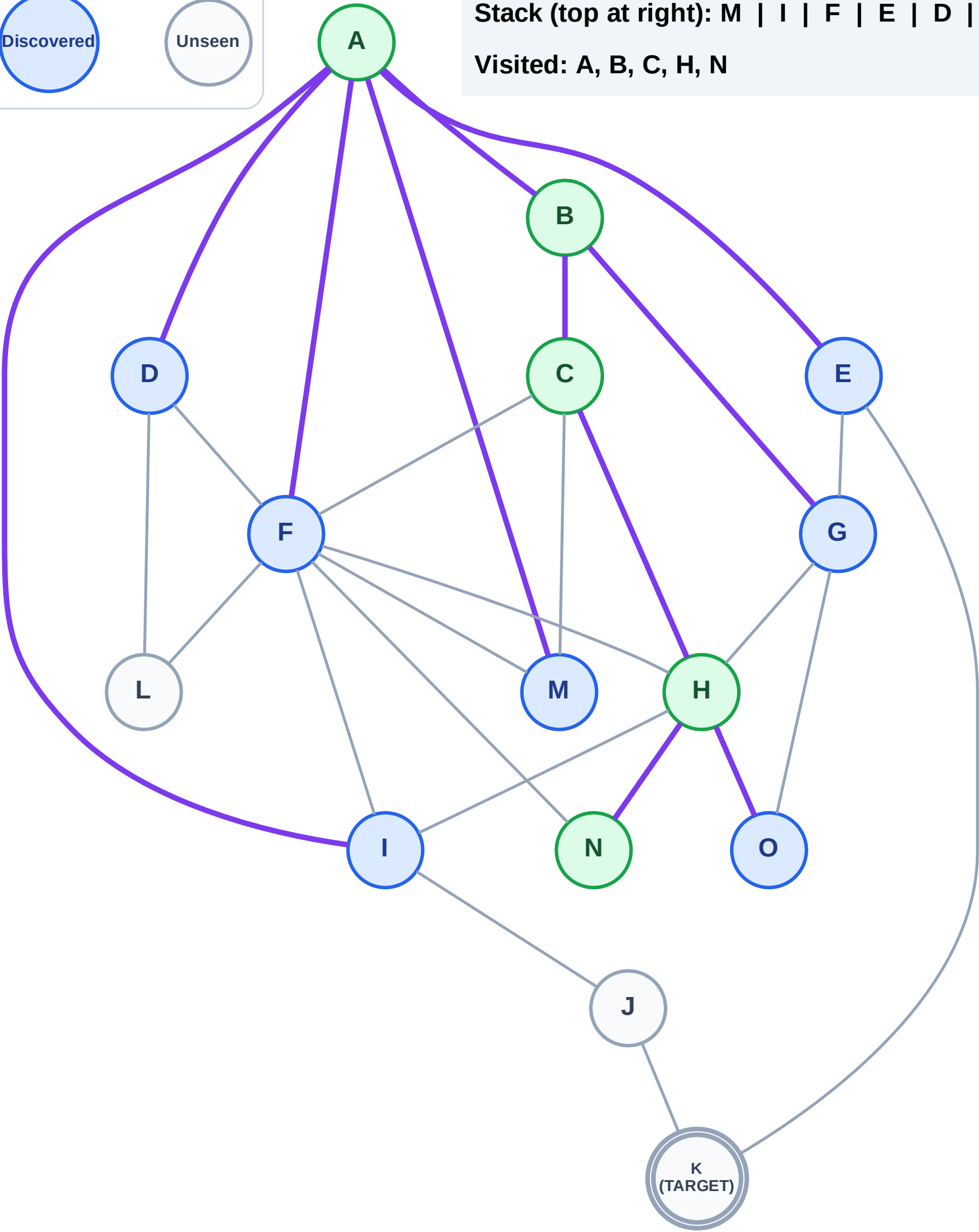
Processing

Discovered

Unseen

Stack (top at right): M | I | F | E | D | G | O

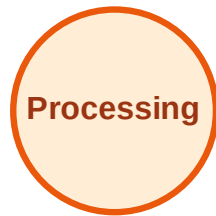
Visited: A, B, C, H, N



DFS Traversal

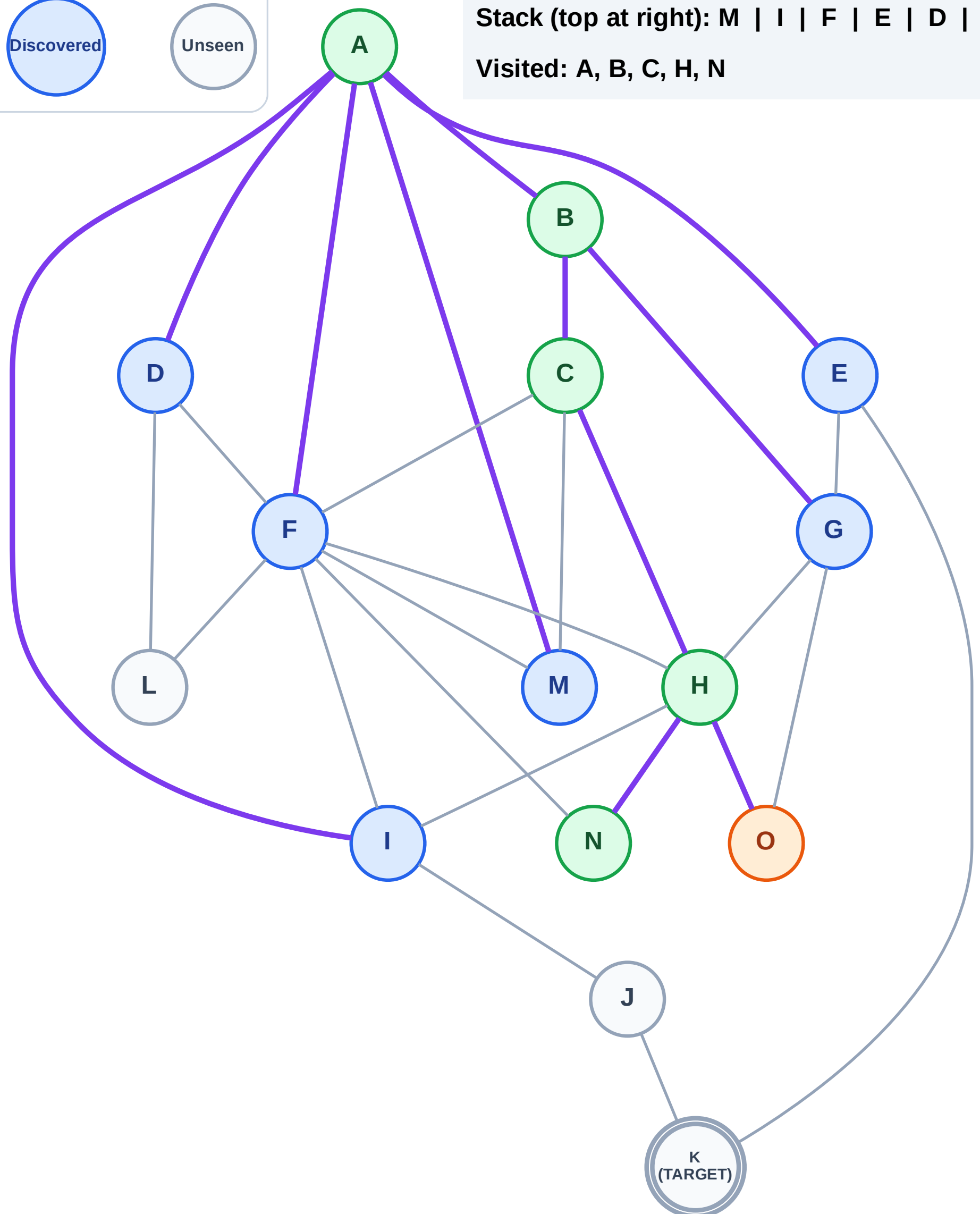
Step 12: Process node O

Legend



Stack (top at right): M | I | F | E | D | G

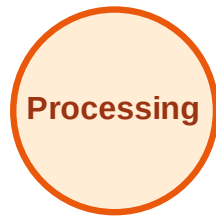
Visited: A, B, C, H, N



DFS Traversal

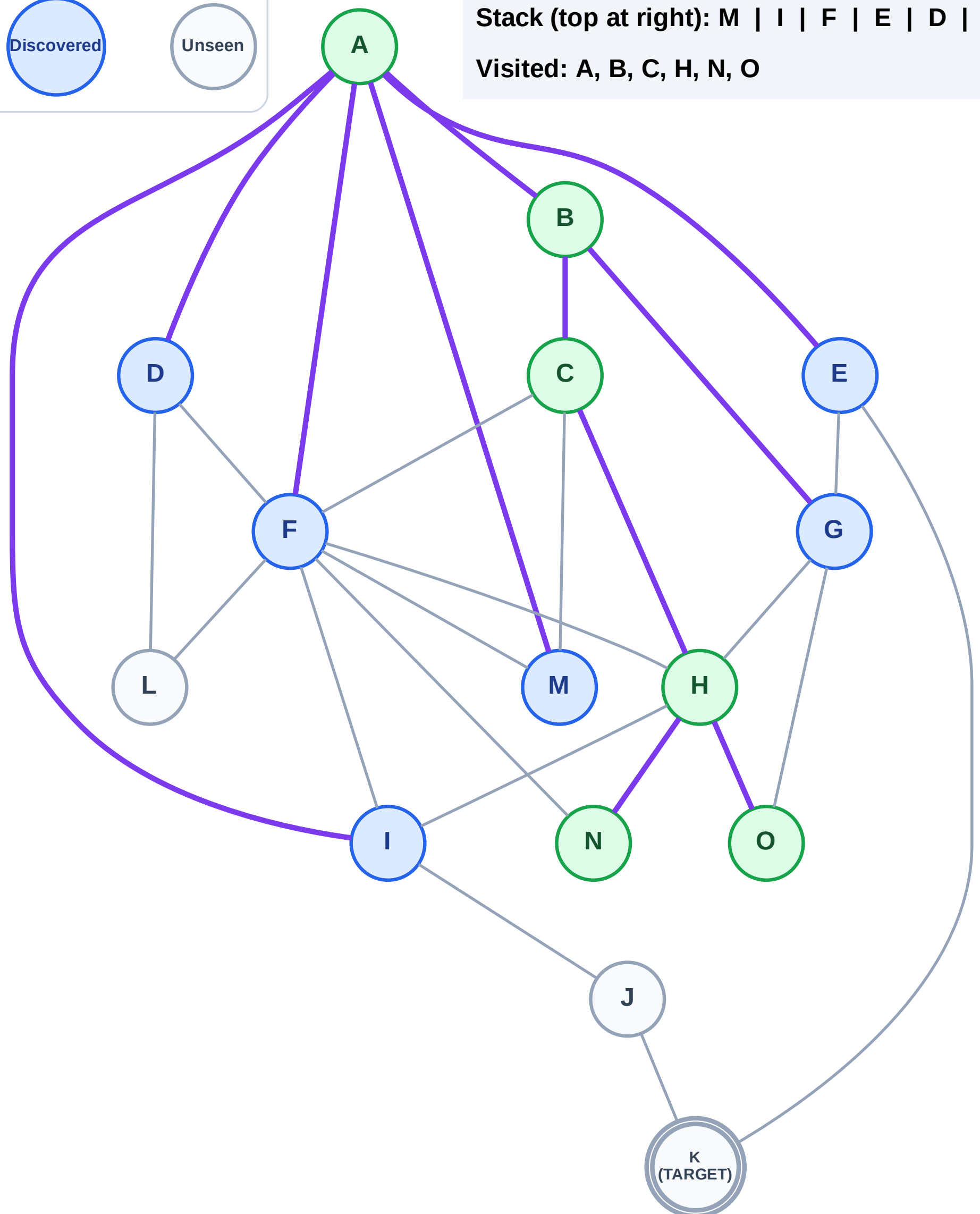
Step 13: No new neighbors from O

Legend



Stack (top at right): M | I | F | E | D | G

Visited: A, B, C, H, N, O



DFS Traversal

Step 14: Process node G

Legend

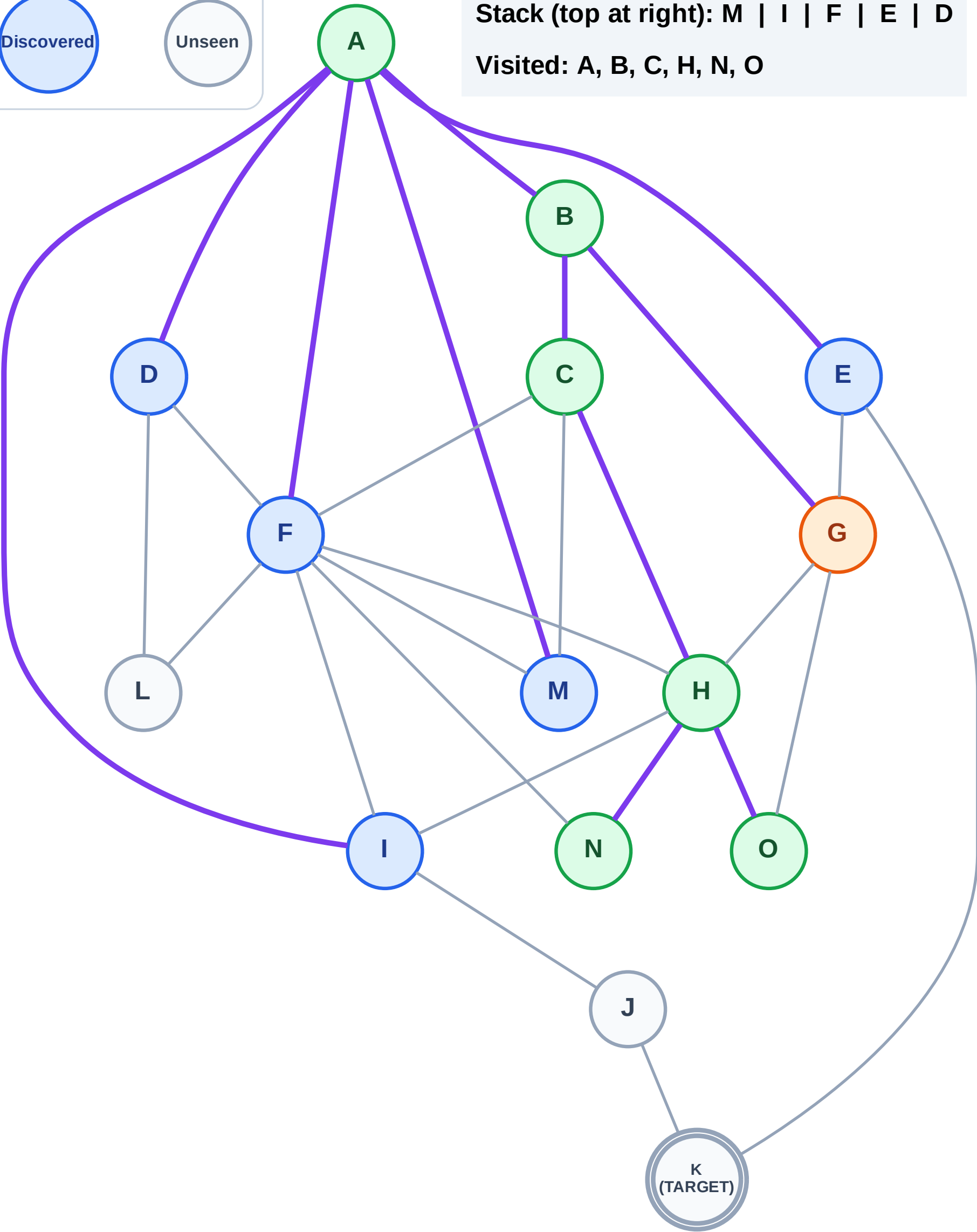
Complete

Processing

Discovered

Unseen

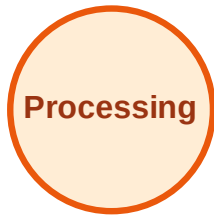
Stack (top at right): M | I | F | E | D
Visited: A, B, C, H, N, O



DFS Traversal

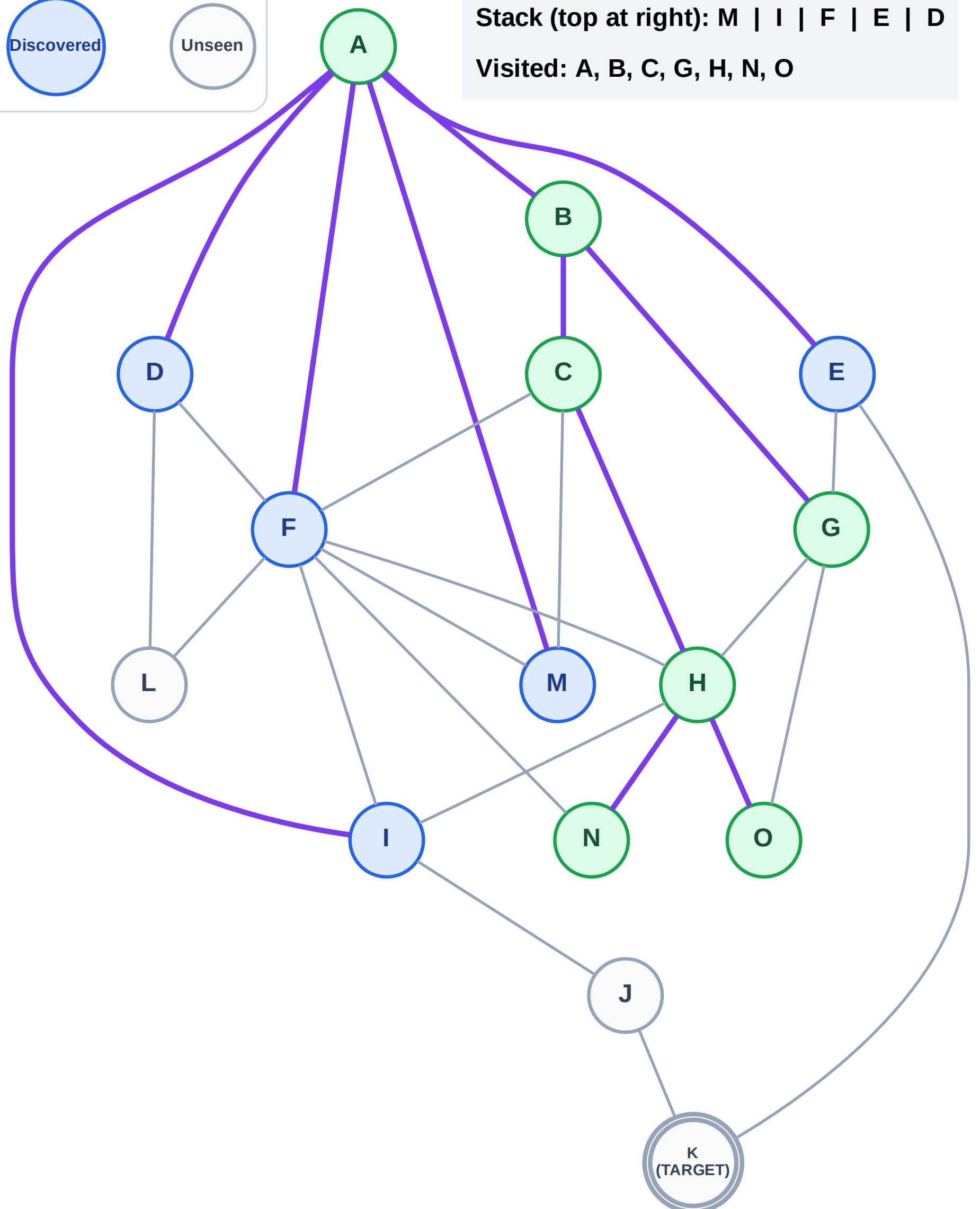
Step 15: No new neighbors from G

Legend



Stack (top at right): M | I | F | E | D

Visited: A, B, C, G, H, N, O



DFS Traversal

Step 16: Process node D

Legend

Complete

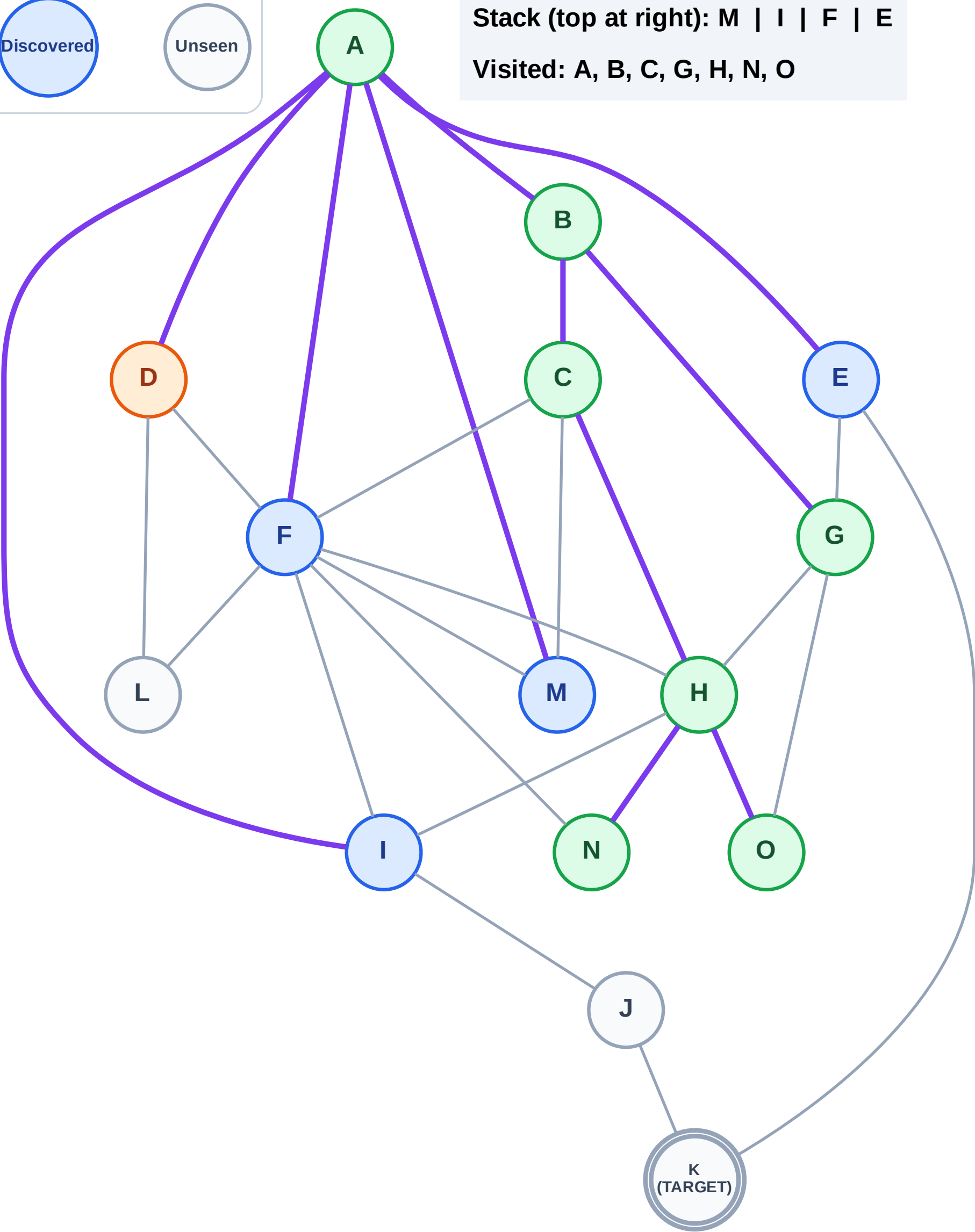
Processing

Discovered

Unseen

Stack (top at right): M | I | F | E

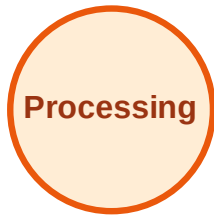
Visited: A, B, C, G, H, N, O



DFS Traversal

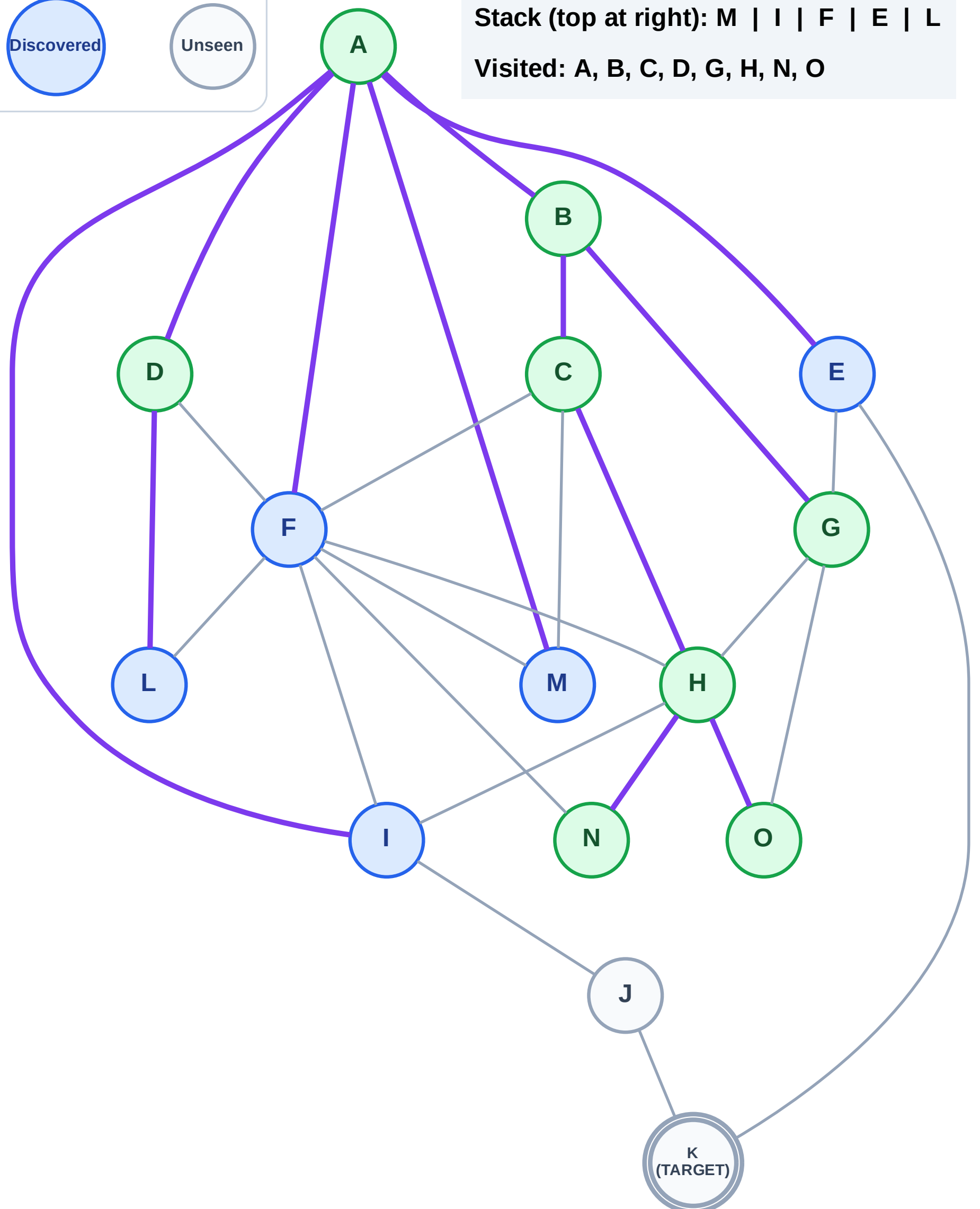
Step 17: Discover L from D

Legend



Stack (top at right): M | I | F | E | L

Visited: A, B, C, D, G, H, N, O



DFS Traversal

Step 18: Process node L

Legend

Complete

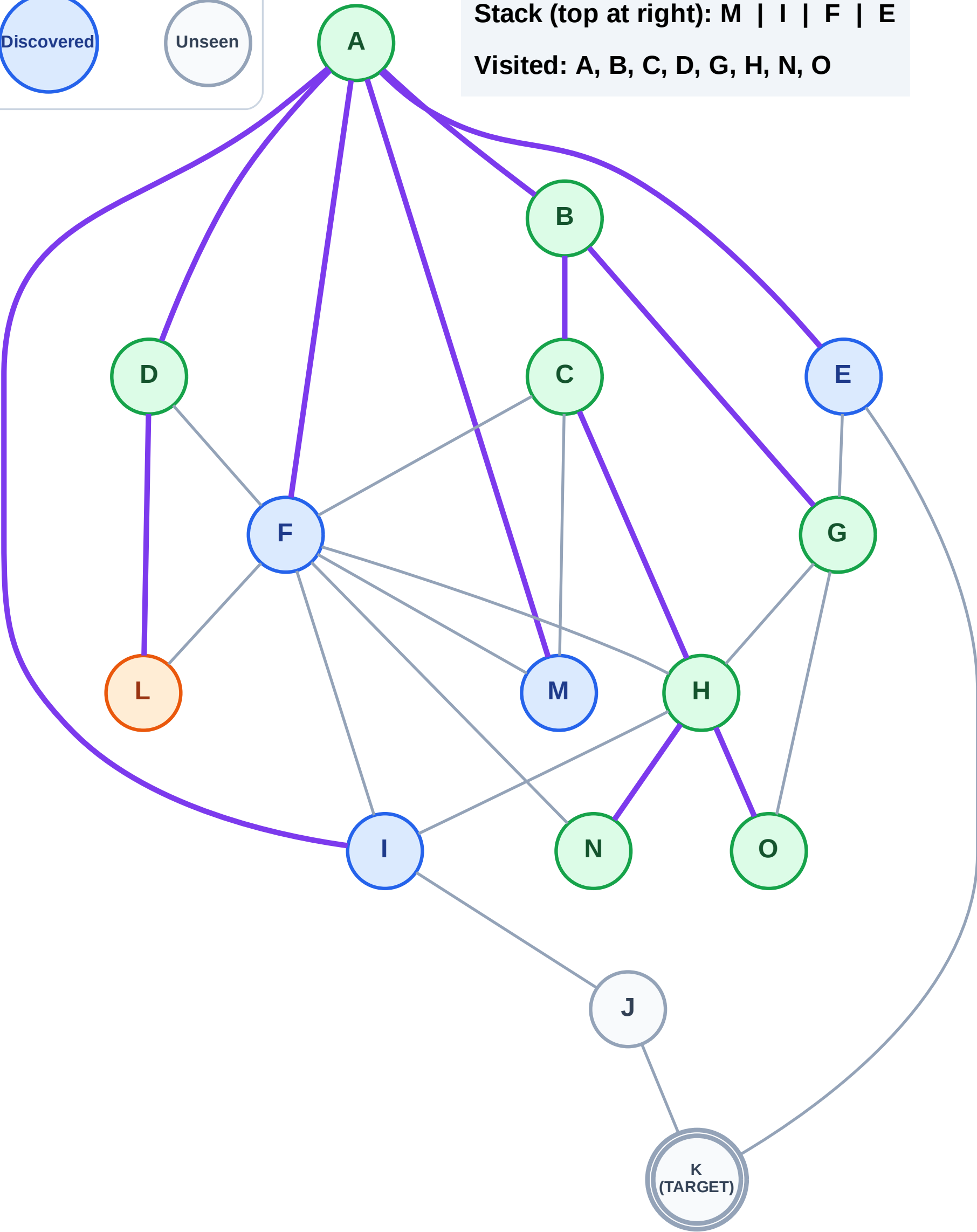
Processing

Discovered

Unseen

Stack (top at right): M | I | F | E

Visited: A, B, C, D, G, H, N, O



DFS Traversal

Step 19: No new neighbors from L

Legend

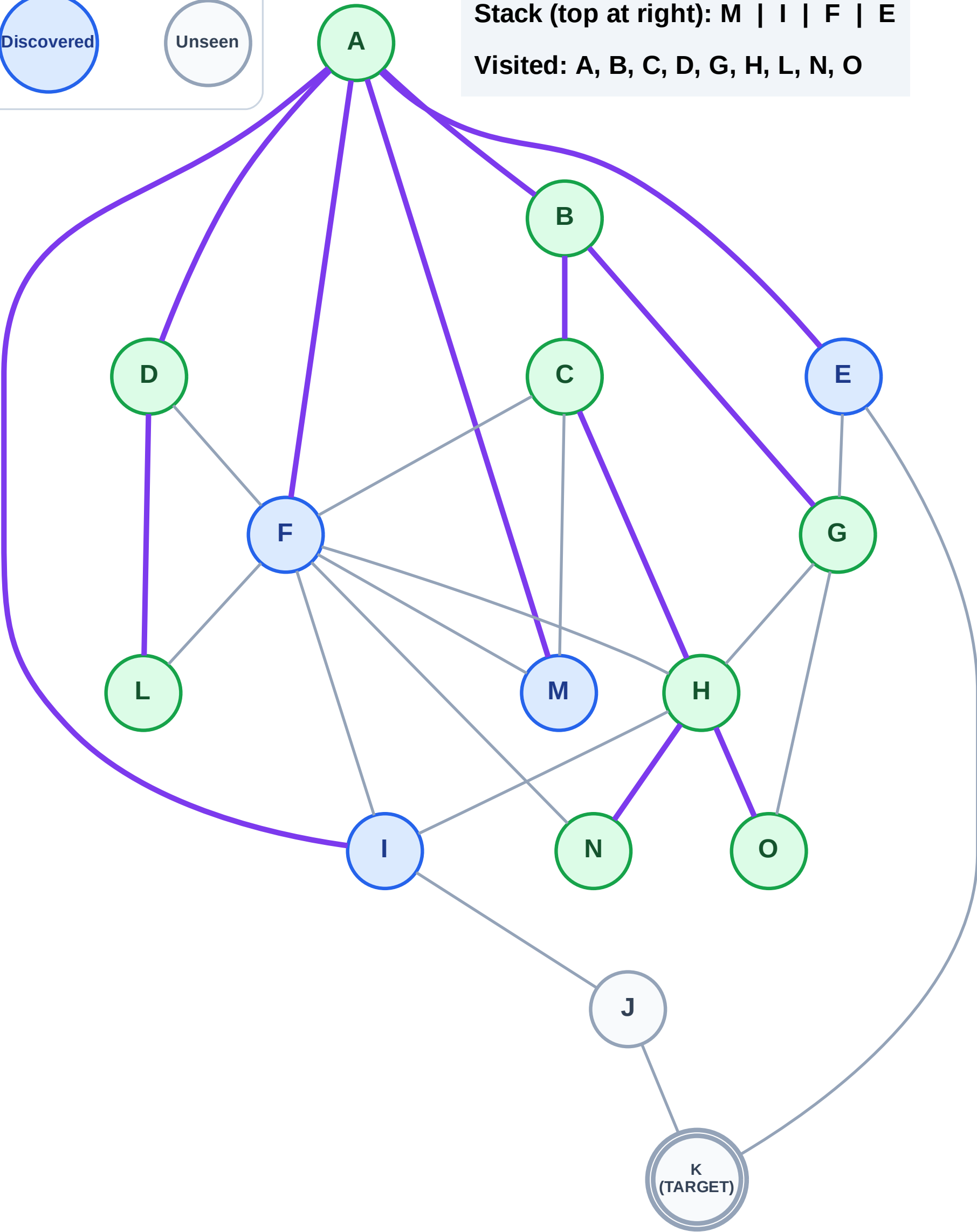
Complete

Processing

Discovered

Unseen

Stack (top at right): M | I | F | E
Visited: A, B, C, D, G, H, L, N, O



DFS Traversal

Step 20: Process node E

Legend

Complete

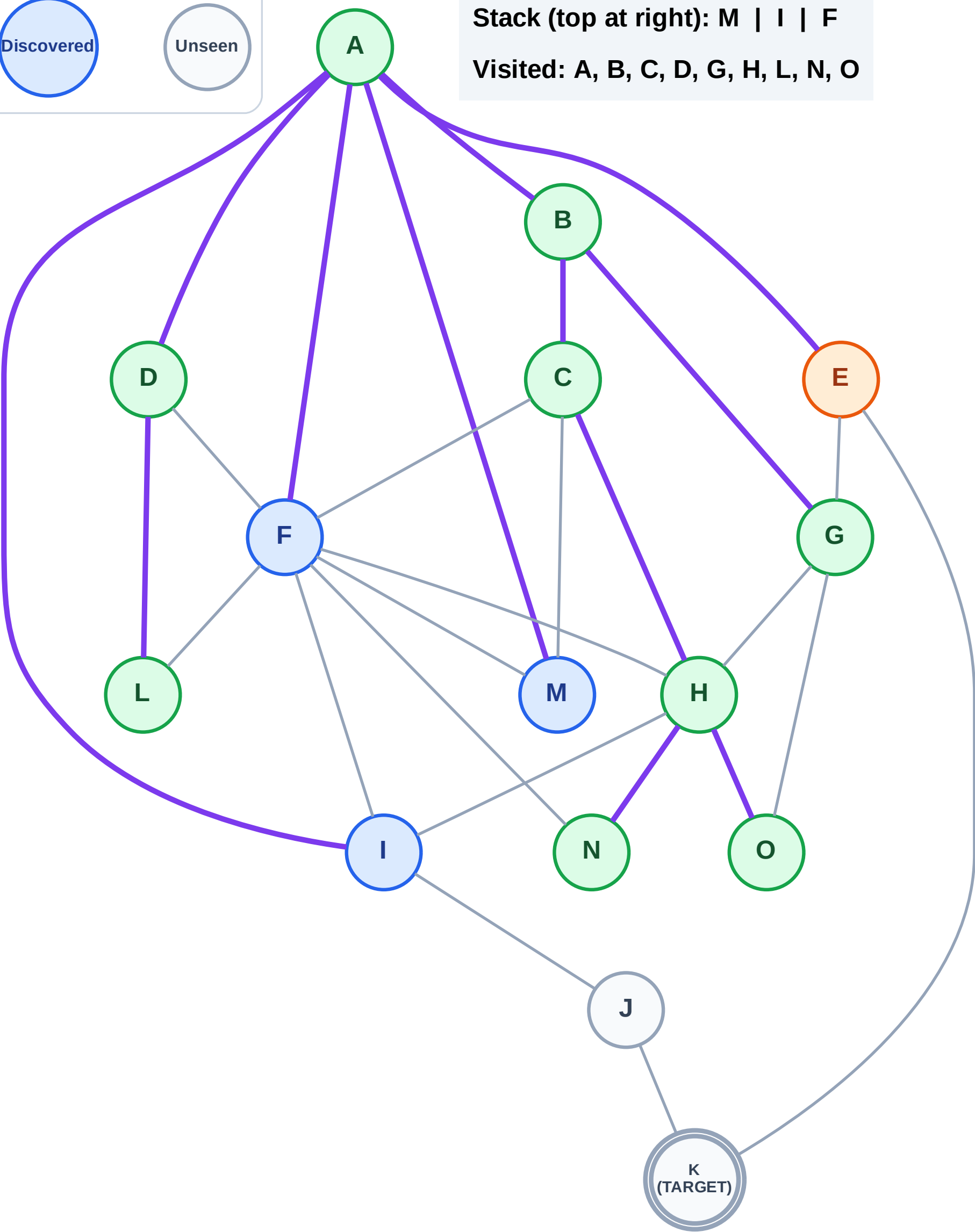
Processing

Discovered

Unseen

Stack (top at right): M | I | F

Visited: A, B, C, D, G, H, L, N, O



DFS Traversal

Step 21: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | I | F

Visited: A, B, C, D, E, G, H, K, L, N, O

